Manuscript AIS 3400076: revision workbook

Every editor and reviewer comment, exact edits in LaTeX, and the response letter

Built from the decision letter (22 Sep 2026), the submitted manuscript and SI, the repository code and the original datasets.

How to use this document

1. Work through the numbered **Steps**. Each gives *where* (manuscript or SI, section, paragraph), the current text to **find** as printed in the submitted PDF, and the LaTeX to paste in a yellow box.
2. Select the text in a yellow box and copy it (the pages have no numbers, so a box that continues on the next page also copies cleanly). Every box is also a file in `snippets/` with the name shown in its title; copying from the file is the safest option.
3. Green boxes are the response-letter text for each comment. They are already assembled, with the reviewer quotes, in `snippets/response_letter.tex`, which compiles as is; `response_letter.pdf` is the compiled version.
4. Grey boxes quote the reviewer, blue boxes say what to do, and red boxes are notes for you only. **[FILL: ...]** marks information only you have; search the snippets for `[FILL` and `[CONFIRM]` before submission.
5. Mark the changes in the revised manuscript (the editor may ask for it), e.g. by wrapping each pasted text in `\textcolor{blue}{...}`.

Symbols in the snippets. The PDF does not expose your macros, so the snippets use plain notation: $\mathbf{l}$ tendon lengths, $\bar{\mathbf{l}}$ their mean, $\mathbf{l}_{\text{enc}}$ encoder reading, $\mathbf{p}$ tip pose, $\mathbf{t}$ tip position, $\mathbf{J}$ Jacobian, f_p^n, f_C^n, N_l^1, N_p^3, N_l^3, N_C networks, W^3 planner, ζ weight of the secondary term, $H(\mathbf{l})$ secondary cost, $\boldsymbol{\xi}$ and $\mathbf{q}$ PCC parameterizations. Replace them with your own macros where they differ.

Numbering of new SI items. So that the existing SI numbers (Tables S1–S6, Figs. S1–S6) stay unchanged, put *all new table and figure environments at the end of the SI* (Step 95), in this order: Tables S7 (path errors), S8 and S9 (targets), S10 (comparison), S11 (reproducibility), S12 (dataset composition), S13 (ablation), S14 (train/test overlap); then Figs. S7 (error vs. bending), S8 (ablation), S9 (coverage). The text snippets refer to these numbers. New SI text goes into the sections named in each step, including two new subsections (5.3 and 6.6).

What was verified. Table 1, Table S6, Fig. 5c, Fig. 8b, Table 2 and the bars of Fig. 7 were reproduced from the repository and the original datasets with the original split. Every notebook seeds Python, NumPy and PyTorch with 42 before splitting. Table S5 is reproduced within 0.02 mm and 0.08°.

Files (in `~/Downloads/AIS_3400076_revision/`):

- `snippets/`: every LaTeX box and the response letter.
- `figures_for_SI/`: the PDFs used by the new SI figures, already named as in the snippets.
- `figures/`: proposed versions of the revised figures.
- `crops/`: annotated crops of the current figures.
- `scripts/` and `data/`: every analysis and its output.
- `originals/`: the review material moved out of the repository.

Checklist: every comment and where it is handled

ID	Comment	What changes	Section
G1–G6	Editor: title, references, acronyms, figure text, copyright, DOIs	Acronym pass, captions, DOIs, statement	3.1
1	Role of visual sensing in control	Abstract, contributions, Sec. 3.3, Alg. 1, Fig. 4c, Conclusion	1.4, 3.3
2	Originality vs. prior work	Intro paragraph, contributions, Table S10, 4 references	2
3	Scalability claim	Sec. 3.1, Intro, contributions, Conclusion	3.2
4	Ground truth of the reconstruction	Assumptions (Sec. 2.2), Sec. 4.2.2, SI 4.2, Fig. S2	3.5, 1.10
5	Arc-length inconsistency	Eqs. 4–6, SI 2.2, SI Eqs. S3–S4, Table S1	3.4
6	Secondary objective	Sec. 3.3.2 wording, SI 5.3, Table S13, Fig. S8	3.8
7	Coverage of the control evaluation	Sec. 3.2 range, Secs. 4.1.4, 4.2.6, SI 6.3, Tables S8–S9, Fig. S9, Sec. 5	3.8
8	Train/test independence	Secs. 3.1.1, 3.2, SI 4.1, Tables S12, S14	3.5
9	Reproducibility	Table S11, SI 3.1, Sec. 5 (lighting)	3.6, 3.7
10	Figures 1B, 2, 3, 5	Part 1	1
11	“Markerless”	Contribution 1	3.2
12	Section vs. segment	Replacement list	3.2
13	Number of tendons, preload	Sec. 2.1	3.7
14	Camera details	Sec. 2.1, Table S11	3.7

ID	Comment	What changes	Section
15	“Simultaneous” acquisition	Sec. 2.1, Fig. 1 caption	3.7, 1.1
16	Sine/cosine, loss balance	SI 3.1	3.6
17	Sign in Eq. S11, highlighted pixels	SI 3.2, Fig. 4 caption	3.6, 1.4
18	PCC line in Alg. 1	Alg. 1, Sec. 3.3.2	3.3
19	Jacobian notation	Secs. 3.3, 3.3.1, Eq. 7, Algs. 1–2, Fig. S3, Table S2	3.3
20	Open-loop baselines, terminology	Sec. 3.3.1, “feedforward”	3.3
21	Percentage denominators	Secs. 4.1.1, 4.1.4, 4.2.1, 4.2.3, 4.2.4	3.8
22	Fig. 5b scaling	Fig. 5b, caption	1.5
23	Fig. 7 statistic	Fig. 7, caption, Secs. 4.1.2, 4.2.2	1.7
24	Table 1 robust statistics	Table 1, Sec. 4.2.1	3.8
25	Bending claim	Fig. S7, Secs. 4.1.1, 4.2.1	3.8
26	Figs. 6, 9, 10 details, path errors	Captions, Table S7	1.6
27	Demonstrated vs. proposed	Sec. 5	3.2
28	Fig. S2b numbers	Fig. S2 caption, SI 4.2	1.10
29	Table S6 skew	Table S6, SI 6.1	3.8
30	Figs. S4–S6 IDs, colors	Figures, captions	1.12
31	Tables S3/S4 composition	SI 4.1, Table S12	3.5

1 Figures and captions

Reviewer, Comment 10

Several figures—particularly Figs. 1B, 2, 3, and 5—are difficult to interpret. [...] In particular, Figure 3 requires attention. [...] The paper should use a consistent flow direction, label all network inputs and outputs, and distinguish clearly between the networks used for pose estimation, curvature reconstruction, forward kinematics, and inverse kinematics.

Summary. Graphics change only where a comment asks for it: Fig. 3 (redraw), Fig. 7 (new statistic) and Figs. S4–S6 (target IDs, shared colors). Fig. 4c gets one added block and Fig. 5b equal axes. All other figures change only in their captions. Three new SI figures are added (S7–S9).

Figure	Change	Type
1	Software synchronization, "sections", every label of panel b	caption
2	L_1–L_4 vs. tendons	caption
3	Network names, inputs/outputs, sizes, one flow direction	redraw + caption
4	Grad-CAM definition (a); encoder path, curvature outside the loop (c)	small edit + caption
5	Equal axis scaling in b; CL/OL and error bars in c	small edit + caption
6, 9, 10	Start, direction, cycles, single run, "open-loop"; path errors	caption + Table S7
7	Per-point error vs. normalized arc length	redraw + caption
8, S1	"section"; CL/OL; error bars	caption
S2	Configurations, interval, independence	caption
S3	Jacobian symbols	small edit
S4–S6	Target IDs, shared color scales	redraw + caption
S7–S9 (new)	Error vs. bending; secondary-objective ablation; coverage	new

1.1 Figure 1

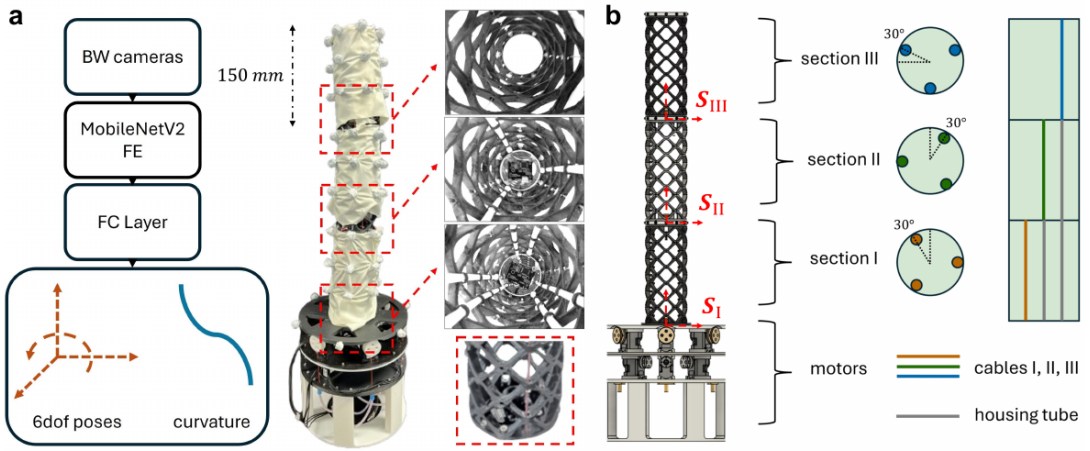


FIGURE 1 | (a) Multi-section soft robot arm sensorization and reconstruction pipeline. Images from the three cameras are acquired simultaneously and fed to CNNs, which predict both the tip poses and the full-body curvature. (b) Robot setup. The tendon-driven arm consists of three independent segments, each actuated by three motors, with Bowden cables routing the tendons of sections II and III. The optical tracking system provides the ground-truth configuration.

Reviewer, minor comments 12 and 15

Use either 'section' or 'segment' consistently [...] The claim that images are acquired simultaneously should be supported by the synchronization method as this type software capture is not necessarily simultaneous hardware triggering.

Step 1 Manuscript, Fig. 1: replace the whole caption.

The graphic stays as it is. The new caption:

- names every element of panel b;
- expands the acronyms of panel a (BW, FE, FC);
- describes the software synchronization (in the code, each camera is read in its own thread and the latest frame is used; there is no hardware trigger).

LaTeX to paste • snippets/fig1_caption.tex

```
\caption{(a) Multi-section soft robot arm sensorization and reconstruction pipeline. Images from the three
cameras (BW: black-and-white, i.e., grayscale) are read concurrently by software, one capture thread per camera,
without hardware triggering (Section 2.1), and fed to CNNs (FE: feature extractor; FC: fully connected layer),
which predict both the section-tip poses and the full-body curvature. (b) Robot setup. The tendon-driven arm
consists of three independent sections, each actuated by three motors, with Bowden cables (housing tubes in gray)
routing the tendons of sections II and III. The circles show the tendon positions in the cross-section of each
section, and $S_{\mathrm{I}}$--$S_{\mathrm{III}}$ the section base frames. The optical tracking system provides
the ground-truth configuration.}
```

For you only (not for the reviewer)

Check that S_{I}–S_{III} in Fig. 1b are the section base frames; if they denote something else, change that half-sentence.

1.2 Figure 2

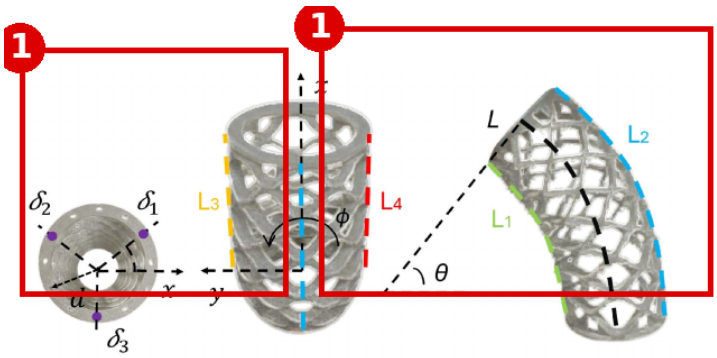


FIGURE 2 | PCC parameterization of soft robots according to both $\alpha\,(\theta, \phi, L)$ and $q\,(\Delta_x, \Delta_y, L)$ parameters, derived from the four arc lengths $\{L_i\}_{i=1,\dots,4}$. The tendon angles $\{\delta_i\}_{i=1,\dots,3}$ and the radius d are also indicated.

Step 2 **Manuscript, Fig. 2: replace the whole caption.**

Graphic unchanged. The confusing point is that four arcs L_1–L_4 are drawn next to three tendons; the caption now says what they are.

LaTeX to paste • snippets/fig2_caption.tex

```
\caption{PCC parameterization of soft robots according to both $\boldsymbol{\xi}(\theta, \phi, L)$ and
$\mathbf{q}(\Delta_x, \Delta_y, L)$ parameters, derived from the four arc lengths $\{L_i\}_{i=1,\dots,4}$.
$L_1$--$L_4$ are the arcs at distance $d$ from the central axis along the $\pm x$ and $\pm y$ directions that
define $\Delta_x$ and $\Delta_y$ (Eq.~S1); they are geometric constructions, not tendons. The three tendons are
placed at the angles $\{\delta_i\}_{i=1,\dots,3}$ on the circle of radius $d$.}
```

1.3 Figure 3

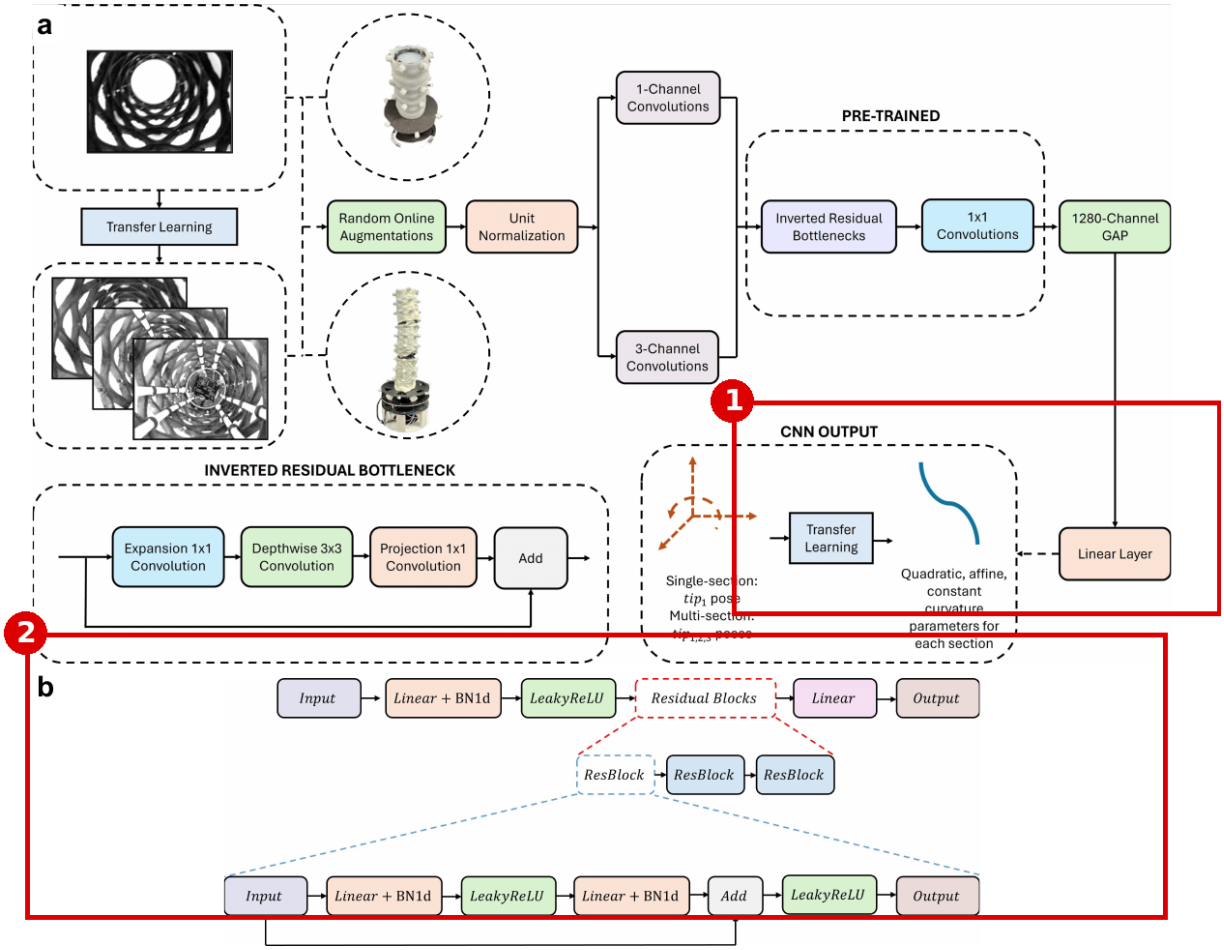


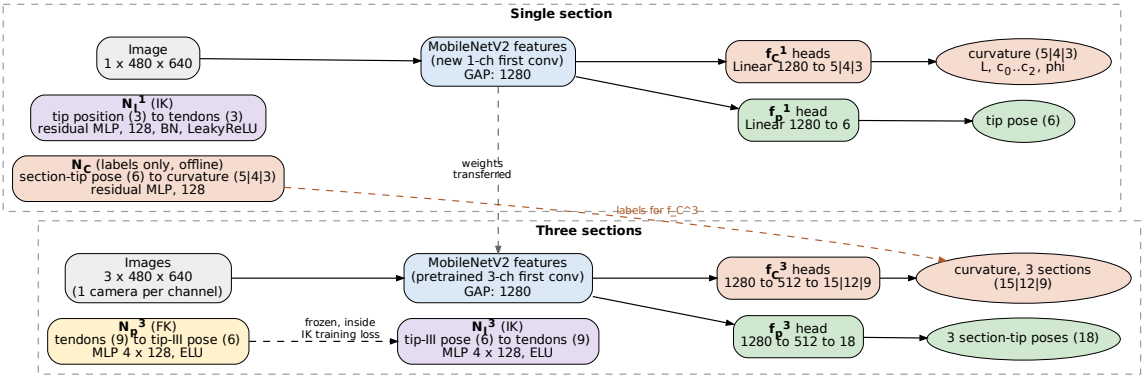
FIGURE 3 | (a) Convolutional neural network (CNN) architecture used for full-body reconstruction. Stacked images from embedded cameras are processed through data augmentation and custom convolutional kernels. A MobileNetV2 backbone with inverted residual bottlenecks is fine-tuned on our dataset and followed by global average pooling (GAP) to produce a compact feature vector. This representation is used to predict tip poses (6 outputs for the single-section, 18 for the multi-section) or curvature parameters for each section, using dedicated networks for different approximations. Transfer learning is applied between pose and curvature networks and to adapt the single-section to the multi-section network. (b) Multi-layer perceptron (MLP) architecture used for kinematics approximations. The network consists of an input projection layer, residual blocks, and an output regression head. The input vector is projected into a 128-dimensional latent space and processed through three residual blocks, each comprising two fully connected layers with Batch Normalization and LeakyReLU activations. Skip connections enable residual learning and help mitigate vanishing gradients.

Step 3 Manuscript, Fig. 3: redraw (explicitly requested).

Keep the two panels and their content, and change:

1. One flow direction, left to right, in both panels.
2. The network name on each output path: f_p^1, f_C^1 (single), f_p^3, f_C^3 (multi) in panel a; N_l^1 and N_C on panel b.
3. Sizes: input $n \times 480 \times 640$; GAP output 1280; heads "1280 $\to$ 6" and "1280 $\to$ 5|4|3" ($n = 1$), "1280 $\to$ 512 $\to$ 18" and "1280 $\to$ 512 $\to$ 15|12|9" ($n = 3$). Marker ①: the single box "Linear Layer" is wrong for $n = 3$, whose heads have two layers.
4. One color per role: pose (green), curvature (orange), forward kinematics (yellow), inverse kinematics (purple).
5. Panel b (marker ②) is the residual MLP of N_l^1 and N_C. Add a small box next to it: "N_p^3: $9 \to 4 \times 128$ ELU $\to 6$; N_l^3: $6 \to 4 \times 128$ ELU $\to 9$, trained through the frozen N_p^3".

Template generated from the code (`figures/network_map.pdf`):



Step 4 **Manuscript, Fig. 3: replace the whole caption.**

LaTeX to paste • `snippets/fig3_caption.tex`

```
\caption{(a) Convolutional neural network (CNN) architecture used for pose estimation and full-body
reconstruction. The $n$ grayscale images from the embedded cameras ($n \times 480 \times 640$) are stacked as
input channels and processed by a MobileNetV2 backbone with inverted residual bottlenecks, fine-tuned on our
dataset and followed by global average pooling (GAP) to a 1280-dimensional feature vector. Regression heads map
this vector to the section-tip poses ($f_p^n$: $6n$ outputs) or to the curvature parameters of each section
($f_C^n$: 5, 4, or 3 parameters per section for the quadratic, affine, and constant models). The heads are one
linear layer for $n = 1$ and two layers ($1280 \to 512 \to$ outputs) for $n = 3$. Data augmentation is applied
during training only. Transfer learning is applied between pose and curvature networks and from the
single-section to the multi-section network. (b) Residual multi-layer perceptron (MLP) used for $N_l^1$ (tip
position, 3 $\to$ tendon lengths, 3) and $N_C$ (section-tip pose, 6 $\to$ curvature parameters of the section).
The input vector is projected into a 128-dimensional latent space and processed through three residual blocks,
each comprising two fully connected layers with Batch Normalization and LeakyReLU activations, followed by the
output regression head. The multi-section kinematic networks $N_p^3$ (tendon lengths, 9 $\to$ tip pose, 6) and
$N_l^3$ (tip pose, 6 $\to$ tendon lengths, 9) are MLPs with four hidden layers of 128 units and ELU activations;
$N_l^3$ is trained through the frozen $N_p^3$ (Eq.~11).}
```

1.4 Figure 4

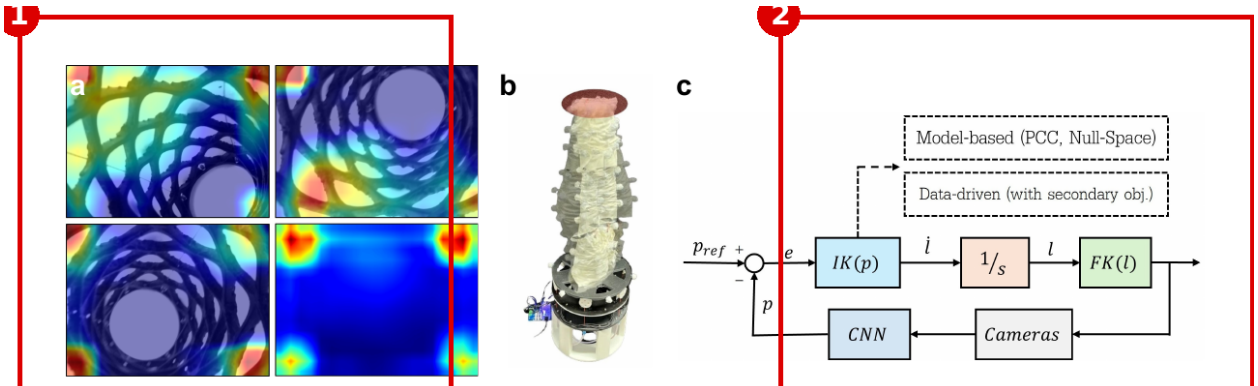


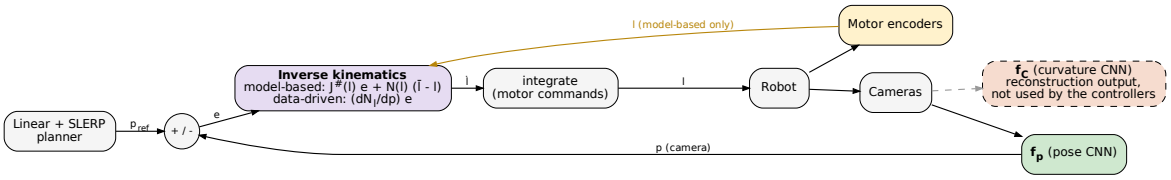
FIGURE 4 | (a) Grad-CAM visualizations from f_p^1 for different bending configurations and average Grad-CAM over the test set. The network mostly focuses on deformations of the Trimmed Helicoid structure along the bending direction. (b) Different null-space configurations yield the same tip pose. (c) High-level kinematic closed-loop controller employing CNN-based feedback. Tendons can be actuated using either model-based or data-driven approaches.

Reviewer, Comment 1 (excerpt) and minor comment 17

The abstract, contribution list, Fig. 4c, and conclusion should clearly distinguish: camera-based pose estimation used for control; encoder measurements used to calculate the Jacobian and null-space term; and image-based PPC reconstruction evaluated as a separate sensing capability. [. . .] clarify how the highlighted pixels are the regions used for prediction.

Step 5 **Manuscript, Fig. 4c: add one block and one arrow (marker ②).**

Add a block “Encoders” that reads $\mathbf{l}$ (the output of the $1/s$ block). Draw an arrow from it to the box “Model-based (PCC, Null-Space)” and label it “$\mathbf{l}$ (model-based only)”. Optionally, add a dashed block “f_C (curvature CNN): reconstruction output, not in the loop”, fed by “Cameras”. Sketch:



Step 6 **Manuscript, Fig. 4: replace the whole caption (covers 4a and 4c).**

LaTeX to paste • `snippets/fig4_caption.tex`

```
\caption{(a) Grad-CAM visualizations from $f_p^1$ for different bending configurations and average Grad-CAM over the test set. Colors show the heatmap of Eq.~S12, normalized to $[0, 1]$ per image and displayed without thresholding; warm colors mark the regions whose activation most reduces the prediction loss. The network mostly focuses on deformations of the Trimmed Helicoid structure along the bending direction. (b) Different null-space configurations yield the same tip pose. (c) High-level kinematic closed-loop controller. The tip pose estimated by the pose CNN from the camera images is the feedback signal of both controllers. In the model-based controller, the motor encoders provide the tendon lengths used by the PCC Jacobian and the null-space term; the data-driven controller does not use encoders. The curvature CNNs are not part of the control loop.}
```

1.5 Figure 5

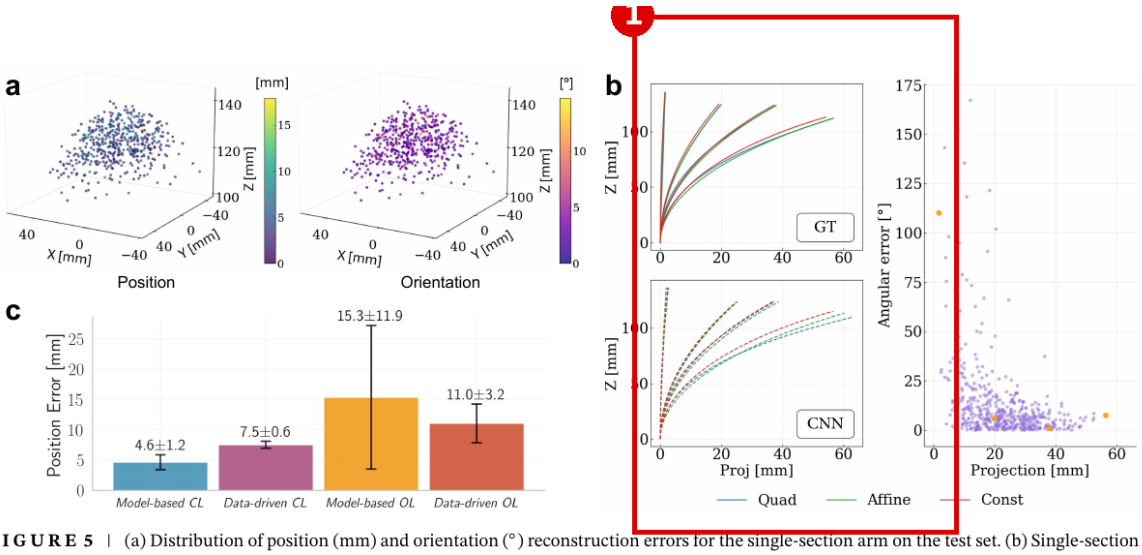


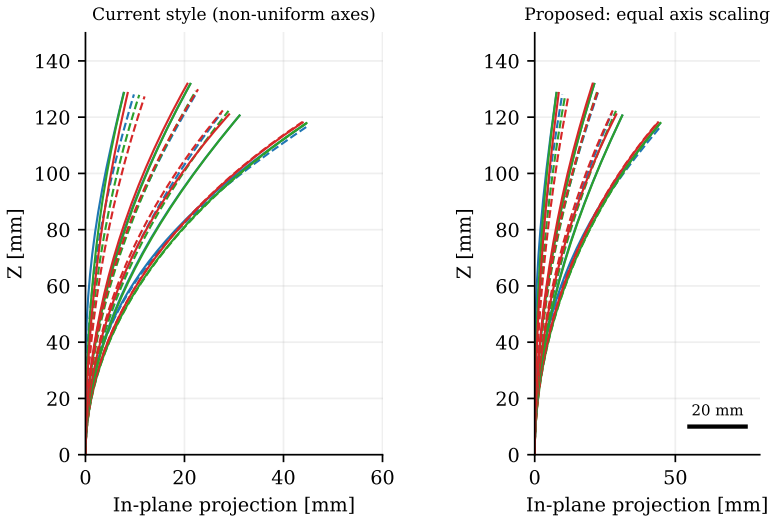
FIGURE 5 | (a) Distribution of position (mm) and orientation (°) reconstruction errors for the single-section arm on the test set. (b) Single-section curvature reconstruction and bending angle error. Four test examples with increasing deformation are shown, with GT (ground-truth) reconstructions as solid and CNN predictions as dashed lines. Axes scaling is not uniform to emphasize the bending effect. The corresponding ϕ angular errors are highlighted in orange. (c) Performance of the proposed controllers on the control workspace of the single-segment arm, reporting both average errors and standard deviations.

Reviewer, minor comment 22

Figure 5b intentionally uses nonuniform axis scaling. This visually distorts the reconstructed shapes. Use equal spatial scaling where possible or add a second undistorted view and explicit scale bars.

Step 7 **Manuscript, Fig. 5b: equal axis scaling (marker ①).**

In the plotting code of the two shape panels (GT and CNN), set `ax.set_aspect('equal')` and give both panels the same limits (e.g. x 0–60 mm, z 0–150 mm). Equal axes make the panels narrow, so the simplest layout is one panel with GT (solid) and CNN (dashed) overlaid. The right panel below shows this layout, produced from the same models (`figures/fig5b_proposed.pdf`):



Step 8 **Manuscript, Fig. 5: replace the whole caption.**

LaTeX to paste • `snippets/fig5_caption.tex`

```
\caption{(a) Distribution of position (mm) and orientation ($^\circ$) reconstruction errors for the
single-section arm on the test set. (b) Single-section curvature reconstruction and bending angle error. Four
test examples with increasing deformation are shown, with GT (ground-truth) reconstructions as solid and CNN
predictions as dashed lines. Axes have equal scaling. The corresponding $\phi$ angular errors are highlighted in
orange. (c) Performance of the proposed controllers on the control workspace of the single-section arm (CL:
closed-loop; OL: open-loop). Bars show the mean final position error over the 20 targets; error bars show one
standard deviation across targets, each visited once.}
```

1.6 Figures 6, 9 and 10

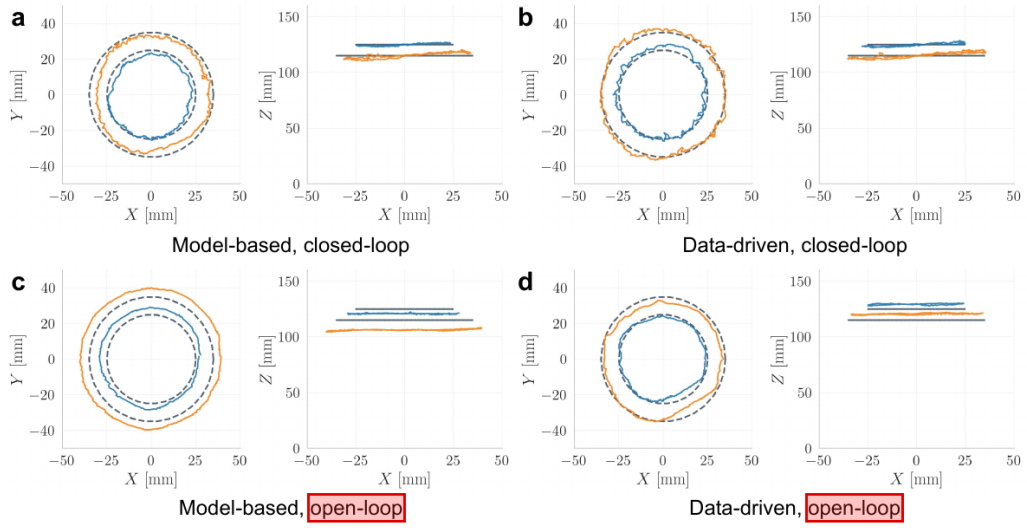


FIGURE 6 | Trajectory tracking for the single-section soft manipulator. The dashed grey line indicates the reference trajectory; executed trajectories are shown in orange for the lower circle and blue for the upper one. (a) Model-based feedback controller. (b) Data-driven feedback controller. (c) Model-based feedforward controller. (d) Data-driven feedforward controller.

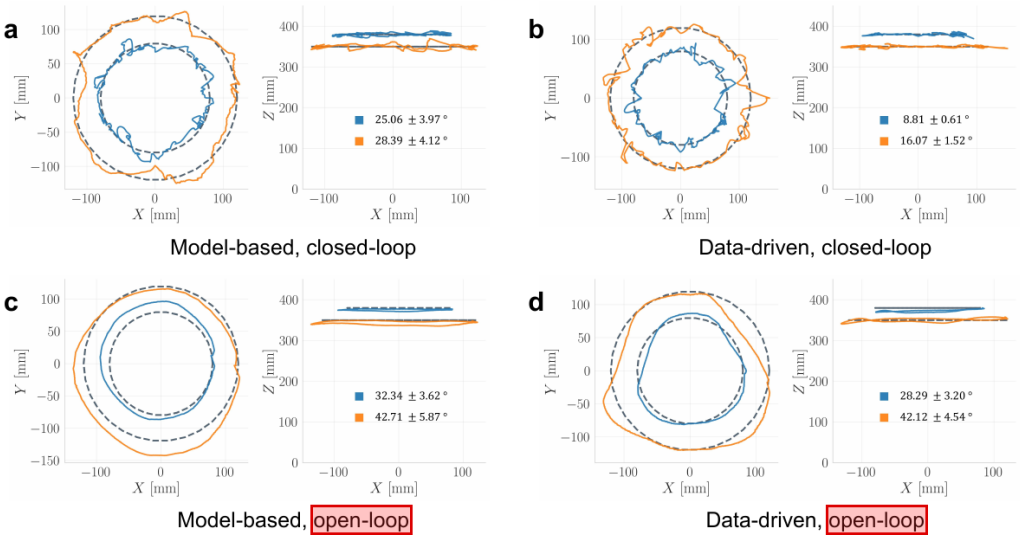


FIGURE 9 | Trajectory tracking for the multi-section soft manipulator. Angular values along both trajectories represent the rotation error on the corresponding circumferences, with smaller values on the inner circle and larger values on the outer one. (a) Model-based feedback controller. (b) Data-driven feedback controller. (c) Model-based feedforward controller. (d) Data-driven feedforward controller.

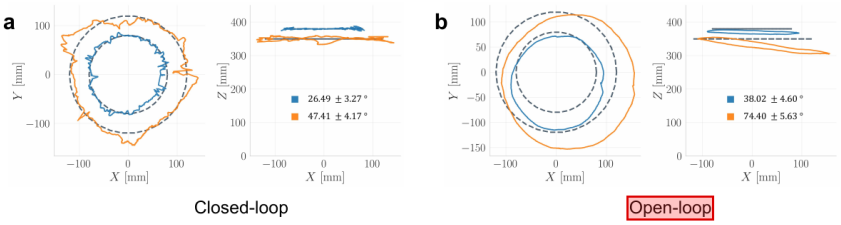


FIGURE 10 | Trajectory tracking for the multi-section robot under an external payload of 50 g attached at the tip, corresponding to about 40% of the effective mass of the manipulator. (a) Model-based feedback controller. (b) Model-based feedforward controller.

Reviewer, minor comments 20 and 26

'Open-loop' and 'feedforward' should be used consistently. [. . .] For Figs. 6, 9, and 10, identify the start direction, waypoint order, number of completed cycles, and whether the plotted paths are one run or an average. Add quantitative path errors in the caption or an accompanying table.

Facts from the logged trajectories:

- Every path starts at angle 0, i.e. at $(r, 0)$ relative to the circle center.
- It visits the 30 vertices counter-clockwise (top view) and completes one cycle.
- Each plotted path is a single run.

The panel titles already say "open-loop"; only the captions say "feedforward".

Step 9 Manuscript, Figs. 6, 9 and 10: replace the three captions.

LaTeX to paste • `snippets/fig6_caption.tex`

```
\caption{Trajectory tracking for the single-section soft manipulator. The dashed gray line indicates the
reference trajectory; executed trajectories are shown in orange for the lower circle and blue for the upper one.
Each trajectory is a single run of one cycle, starting at $(x, y) = (r, 0)$ with respect to the circle center and
traversing the 30 waypoints counter-clockwise (top view); path errors are reported in Table~S7. (a) Model-based
closed-loop controller. (b) Data-driven closed-loop controller. (c) Model-based open-loop controller. (d)
Data-driven open-loop controller.}
```

LaTeX to paste • `snippets/fig9_caption.tex`

```
\caption{Trajectory tracking for the multi-section soft manipulator with horizontal end-effector orientation. The
angular values are the mean $\pm$ standard deviation of the orientation error along each circle and are reported
next to the corresponding circle. Each trajectory is a single run of one cycle, starting at $(x, y) = (r, 0)$
with respect to the circle center and traversing the 30 waypoints counter-clockwise (top view); path errors are
reported in Table~S7. (a) Model-based closed-loop controller. (b) Data-driven closed-loop controller. (c)
Model-based open-loop controller. (d) Data-driven open-loop controller.}
```

LaTeX to paste • snippets/fig10_caption.tex

```
\caption{Trajectory tracking for the multi-section robot under an external payload of 50~g attached at the tip, corresponding to about 40\% of the effective mass of the manipulator. Each trajectory is a single run of one cycle, starting at $(x, y) = (r, 0)$ with respect to the circle center and traversing the 30 waypoints counter-clockwise (top view); path errors are reported in Table~S7. (a) Model-based closed-loop controller. (b) Model-based open-loop controller.}
```

Step 10 **SI, end of the document: add Table S7 (path errors, generated from the repository's `PathErrorMetrics` files).**

LaTeX to paste • snippets/tab_S7_paths.tex

```
\begin{table}[h]
\centering
\caption{Path errors of the trajectory-tracking experiments (Figs.~6, 9 and 10). Radial RMS: root-mean-square distance of the executed path from the reference circle, measured in the horizontal plane; height RMS: root-mean-square of $z - z_{\mathrm{ref}}$. The reference circle is the commanded one (center and radius of the commanded waypoints). Each row is a single run of one cycle; consecutive samples along a path are serially correlated and are not independent trials.}
\label{tab:S7_paths}
\begin{tabular}{lllrrr}
\toprule
Arm & Controller & Loop & $z_{\mathrm{ref}}$ [mm] & Radial RMS [mm] & Height RMS [mm] \\
\midrule
Single & Model-based & closed & 115 & 1.46 & 2.65 \\
Single & Model-based & closed & 125 & 1.32 & 1.23 \\
Single & Data-driven & closed & 115 & 1.71 & 2.49 \\
Single & Data-driven & closed & 125 & 1.83 & 1.54 \\
Single & Model-based & open & 115 & 8.06 & 8.74 \\
Single & Model-based & open & 125 & 5.71 & 4.30 \\
Single & Data-driven & open & 115 & 2.27 & 5.48 \\
Single & Data-driven & open & 125 & 1.61 & 4.41 \\
\midrule
Multi & Model-based & closed & 350 & 7.78 & 5.74 \\
Multi & Model-based & closed & 380 & 6.25 & 3.43 \\
Multi & Data-driven & closed & 350 & 7.63 & 2.10 \\
Multi & Data-driven & closed & 380 & 5.85 & 2.72 \\
Multi & Model-based & open & 350 & 12.64 & 8.10 \\
Multi & Model-based & open & 380 & 10.60 & 5.24 \\
Multi & Data-driven & open & 350 & 10.86 & 4.41 \\
Multi & Data-driven & open & 380 & 5.53 & 7.67 \\
Multi & Model-based, 50 g & closed & 350 & 11.46 & 6.46 \\
Multi & Model-based, 50 g & closed & 380 & 5.35 & 1.63 \\
Multi & Model-based, 50 g & open & 350 & 27.10 & 28.46 \\
Multi & Model-based, 50 g & open & 380 & 20.64 & 11.22 \\
\bottomrule
\end{tabular}
\end{table}
```

Preview (how it compiles)

Path errors of the trajectory-tracking experiments (Figs. 6, 9 and 10). Radial RMS: root-mean-square distance of the executed path from the reference circle, measured in the horizontal plane; height RMS: root-mean-square of $z - z_{\mathrm{ref}}$. The reference circle is the commanded one (center and radius of the commanded waypoints). Each row is a single run of one cycle; consecutive samples along a path are serially correlated and are not independent trials.

Arm	Controller	Loop	z_{ref} [mm]	Radial RMS [mm]	Height RMS [mm]
Single	Model-based	closed	115	1.46	2.65
Single	Model-based	closed	125	1.32	1.23
Single	Data-driven	closed	115	1.71	2.49
Single	Data-driven	closed	125	1.83	1.54
Single	Model-based	open	115	8.06	8.74
Single	Model-based	open	125	5.71	4.30
Single	Data-driven	open	115	2.27	5.48
Single	Data-driven	open	125	1.61	4.41
Multi	Model-based	closed	350	7.78	5.74
Multi	Model-based	closed	380	6.25	3.43
Multi	Data-driven	closed	350	7.63	2.10
Multi	Data-driven	closed	380	5.85	2.72
Multi	Model-based	open	350	12.64	8.10
Multi	Model-based	open	380	10.60	5.24
Multi	Data-driven	open	350	10.86	4.41
Multi	Data-driven	open	380	5.53	7.67
Multi	Model-based, 50 g	closed	350	11.46	6.46
Multi	Model-based, 50 g	closed	380	5.35	1.63
Multi	Model-based, 50 g	open	350	27.10	28.46
Multi	Model-based, 50 g	open	380	20.64	11.22

1.7 Figure 7

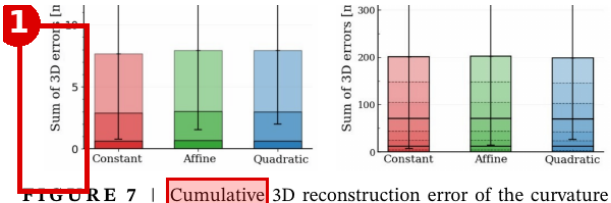


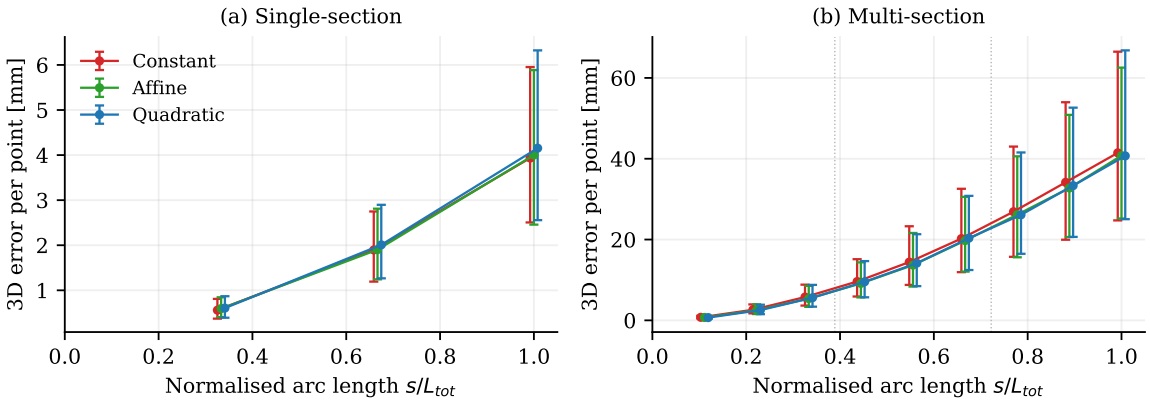
FIGURE 7 | Cumulative 3D reconstruction error of the curvature models. (a) Single-section arm: reference points at $\frac{L}{3}$, $\frac{2L}{3}$, and L. (b) Multi-section arm: reference points placed every ninth of the manipulator length. Error bars indicate ± one standard deviation. Note the different vertical scales.

Reviewer, minor comment 23

Figure 7 alternates among 'cumulative 3D error,' 'sum of 3D errors,' and 'average 3D error for each point.' Define the statistic exactly. Summing three points for the single-section arm and nine for the multi-section arm also prevents direct comparison. Per-point RMSE or error versus normalized arc length would be clearer.

Step 11 **Manuscript, Fig. 7: replace the graphic with the per-point version.**

The new version uses the same models and the same test set. The bar heights of the current figure (7.8, 7.9, 8.3 mm and 209, 203, 198 mm) are the sums of the per-point means shown here. Use `figures/fig7_proposed.pdf` as is, or replot it in your style (median and interquartile range against s/L_{tot}):



Step 12 **Manuscript, Fig. 7: replace the whole caption.**

LaTeX to paste • snippets/fig7_caption.tex

```
\caption{3D reconstruction error of the curvature models along the backbone. For each test sample ($N = 500$),
the backbone reconstructed from the predicted curvature parameters is compared with the one reconstructed from
the curvature labels, and the 3D distance is evaluated at each reference point. (a) Single-section arm: reference
points at $L/3$, $2L/3$, and $L$. (b) Multi-section arm: reference points every ninth of the arm length; the
labels are model-constrained (Section~2.2). The horizontal axis is the arc length normalized by the total length;
markers show the median and error bars the interquartile range.}
```

Step 13 Manuscript, Section 4.1.2, last two sentences: replace.

Find: "Fig. 7a reports the average 3D error for each point, as well as the cumulative standard deviations across all curvature models. Although mean values are comparable, the standard deviation decreases with increasing model descriptiveness."

LaTeX to paste • snippets/text_412_fig7.tex

```
Fig.~7a reports the 3D error at each reference point. The three curvature models give similar errors, which grow
along the backbone to a mean of 4.9--5.1~mm at the tip (median 3.9--4.2~mm).
```

Step 14 Manuscript, Section 4.2.2, last sentence: replace.

Find: "Errors grow toward the distal tip; all models provide similar approximations, though increasing model complexity reduces variance."

LaTeX to paste • snippets/text_422_fig7.tex

```
Errors grow toward the distal tip, reaching a median of about 41~mm at the end of the arm, and all models provide
similar approximations.
```

For you only (not for the reviewer)

Why the standard-deviation sentences go: recomputed on the test set, the spread does not decrease with model complexity. The standard deviation of the summed error is 6.55, 6.27 and 7.34 mm (constant, affine, quadratic) on the single section and 183, 188 and 171 mm on the multi-section arm. The new sentences only state what the per-point figure shows.

1.8 Figure 8

Step 15 Manuscript, Fig. 8: replace the whole caption.

LaTeX to paste • snippets/fig8_caption.tex

```
\caption{(a) Distribution of position (mm) and orientation ($^\circ$) reconstruction errors for the multi-section
arm on the test set. (b) Performance of the four proposed controllers on the control workspace of the
multi-section arm (CL: closed-loop; OL: open-loop), reporting the mean and standard deviation across the 20
targets (each visited once) for translation and rotation accuracy.}
```

1.9 Figure S1

Step 16 SI, Fig. S1: replace the whole caption ("multi-segment", "ends").

LaTeX to paste • snippets/figS1_caption.tex

```
\caption{(a) Workspace of the multi-section robot. Sampled positions of the first, second, and third section tips
are depicted as blue, green, and red dots. (b) Main reference systems during modeling and data acquisition.}
```

1.10 Figure S2

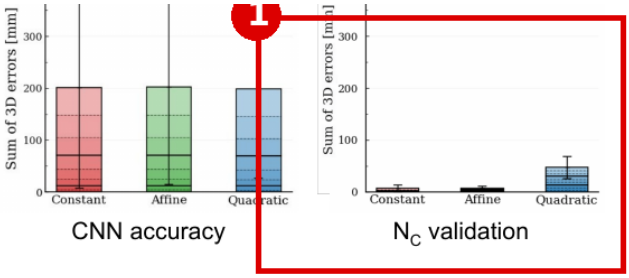


FIGURE S2 | (a) 3D reconstruction accuracy of the curvature models at reference points placed every ninth of the manipulator length. (b) Errors induced by the N_C approximation with respect to the motion-capture ground-truth.

Reviewer, Comment 4 (excerpt) and SI comment 28

Fig. S2b does not provide sufficient validation of N_C: numerical errors, sample count, data split, and confidence intervals are missing. It is also unclear whether its validation configurations are independent of the data used to train N_C and the three-section curvature estimator. [. . .] Fig. S2b should report the numerical N_C error for each curvature model, the number of validation configurations, and whether the bars show standard deviation, standard error, or another interval.

What the code shows:

- There are 20 configurations (`MULTI_curvature_checks.py`), posed manually with all markers visible and recorded in a separate session.
- Each bar is the sum over the nine points of the mean error per point.
- The error bars are the sample standard deviation (ddof = 1) of the per-configuration sum.
- N_C was trained on single-section data only.

Step 17 SI, Fig. S2: replace the whole caption.

LaTeX to paste • `snippets/figS2_caption.tex`

```
\caption{(a) 3D reconstruction accuracy of the curvature models at reference points placed every ninth of the
manipulator length: sum over the nine points of the mean error at each point (same data as Fig.~7b). (b) Errors
induced by the $N_C$ approximation with respect to the motion-capture ground truth, on 20 configurations of the
multi-section arm posed manually with all subsection markers visible and recorded in a separate session; these
configurations were not used to train $N_C$ or the multi-section curvature CNNs. Bars show the sum over the nine
reference points of the mean 3D error at each point; error bars show one standard deviation of the
per-configuration sum across the 20 configurations.}
```

The N_C numbers go into SI Section 4.2 (Step 62).

For you only (not for the reviewer)

Read note 1 in Section 3.11 before finalizing anything that cites values of Fig. S2b.

1.11 Figure S3

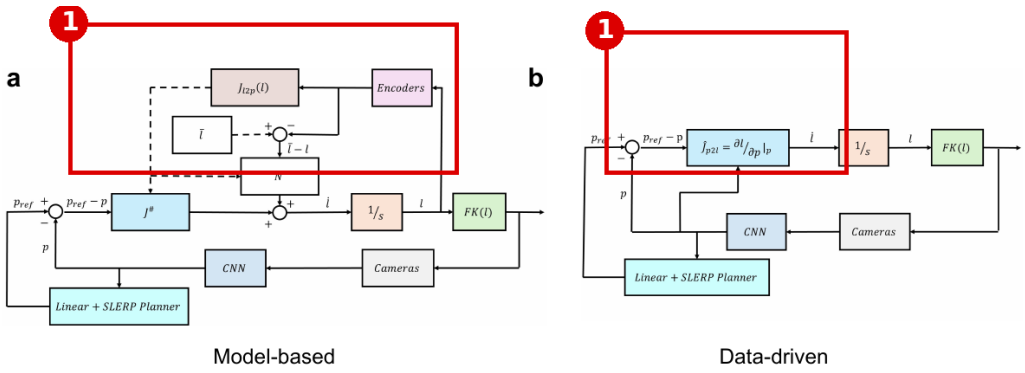


FIGURE S3 | (a) Model-based control scheme for the multi-section soft manipulator based on camera feedback. This approach uses model-based null-space optimization to translate observed poses into actuator commands. (b) Data-driven control scheme for the multi-section soft manipulator based on camera feedback. This approach uses data-driven kinematics learning to translate observed poses into actuator commands.

Step 18 SI, Fig. S3: change the symbols in the blocks (marker ①).

Model-based panel: the Jacobian block becomes $\mathbf{J}^{\#}$ of $\mathbf{J}_{l\to p}(\mathbf{l})$. Data-driven panel: $\hat{\mathbf{J}}_{p\to l}$. This is the notation of Section 3.3. Caption unchanged.

1.12 Figures S4–S6

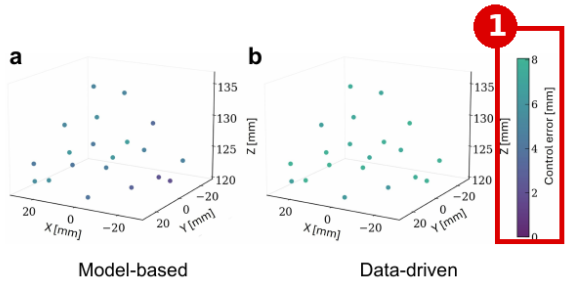


FIGURE S4 | Closed-loop tracking error distribution (mm) on the control workspace for the single-section arm. (a) Model-based. (b) Data-driven.

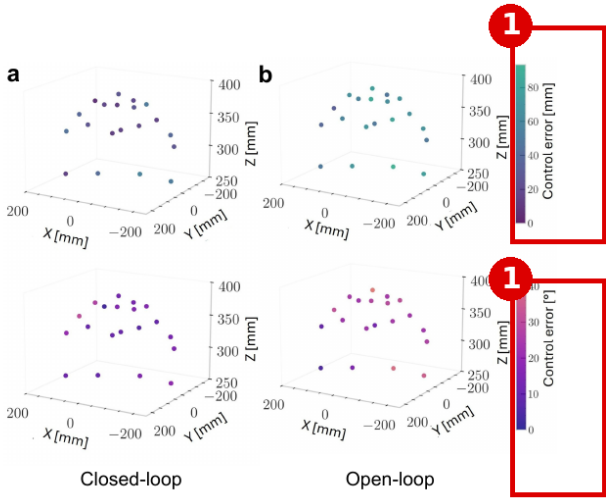


FIGURE S6 | Tracking error distribution (mm and °) for the multi-section arm under external disturbance. (a) Closed-loop. (b) Open-loop.

Reviewer, SI comment 30

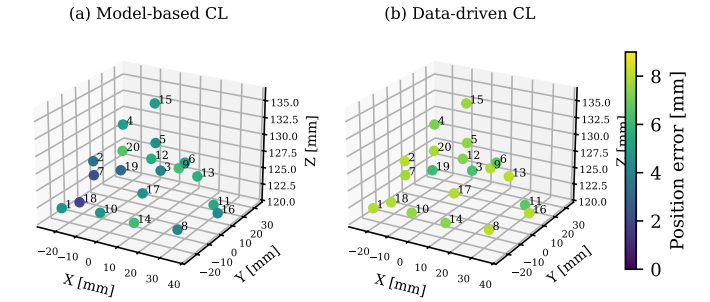
Figs. S4–S6 should use matched target identifiers or connecting pairs so the reader can compare the same workspace locations across controllers. Use consistent color limits when the goal is direct visual comparison.

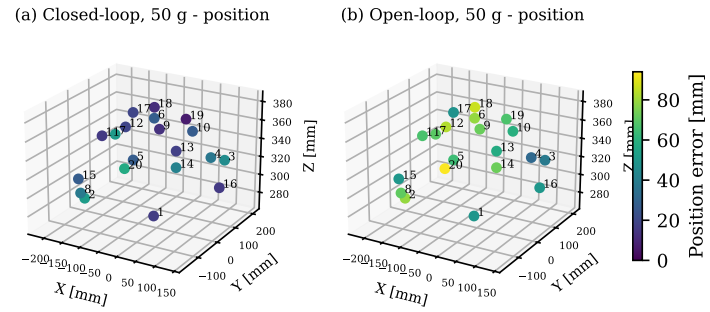
Step 19 SI, Figs. S4–S6: replace the graphics.

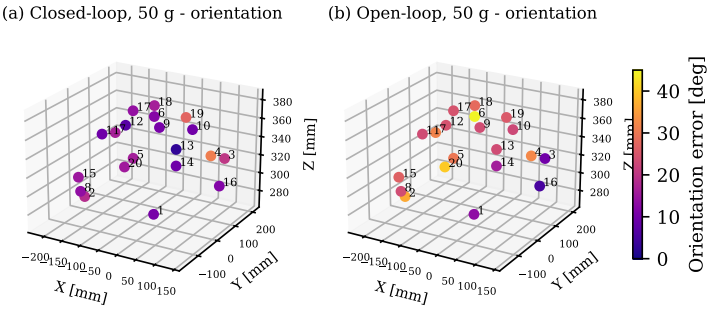
Same data and plots, with two changes:

- Each target carries its ID (1–20, the same as in Tables S8 and S9).
- Each figure uses one color scale per quantity; marker ① shows the current independent color bars.

Ready to use: `figures/figS4_proposed.pdf`, `figS5_proposed.pdf`, `figS6_proposed.pdf`.







Step 20 **SI, Figs. S4, S5 and S6: replace the three captions.**

LaTeX to paste • `snippets/figS4_caption.tex`

```
\caption{Closed-loop tracking error distribution (mm) on the control workspace for the single-section arm. (a) Model-based. (b) Data-driven. Numbers identify the targets (Table~S8); both panels share the same color scale.}
```

LaTeX to paste • `snippets/figS5_caption.tex`

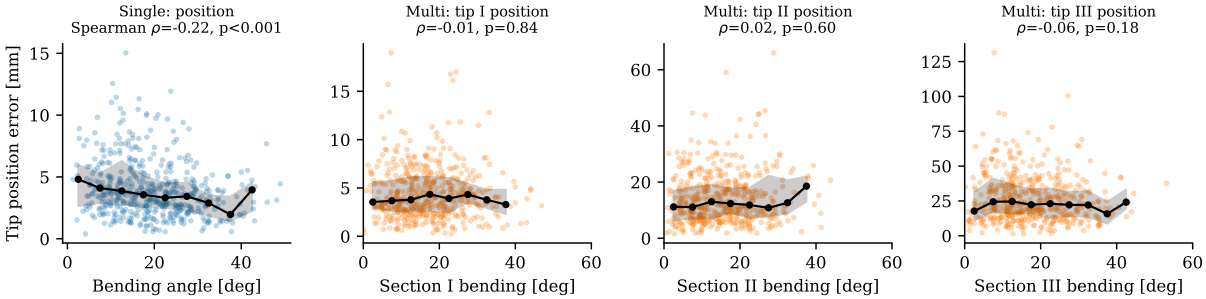
```
\caption{Closed-loop tracking error distribution (mm and $^\circ$) on the control workspace for the multi-section arm. (a) Model-based. (b) Data-driven. Numbers identify the targets (Table~S9); panels showing the same quantity share the same color scale.}
```

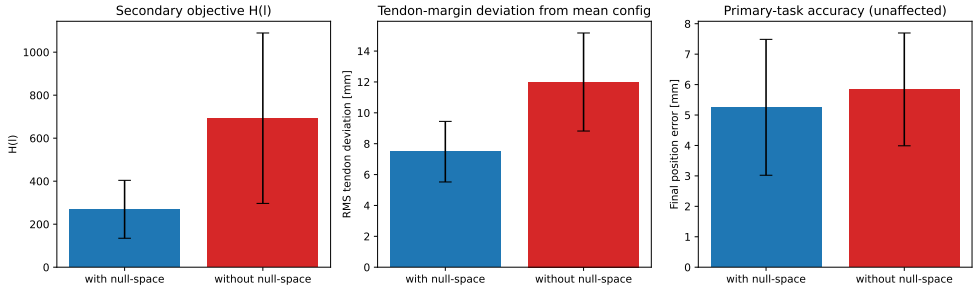
LaTeX to paste • `snippets/figS6_caption.tex`

```
\caption{Tracking error distribution (mm and $^\circ$) for the multi-section arm under external disturbance (50~g tip payload). (a) Closed-loop. (b) Open-loop. Numbers identify the targets (Table~S9); panels showing the same quantity share the same color scale.}
```

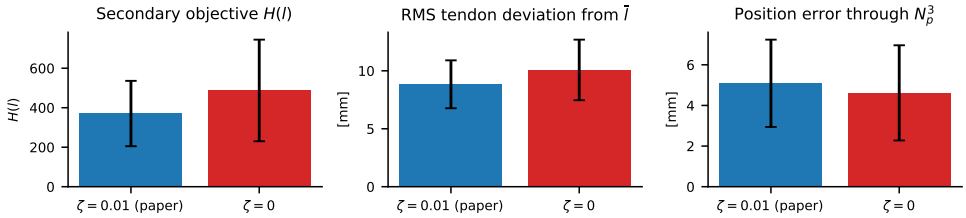
1.13 New SI figures S7–S9

All three go at the end of the SI (Step 95). Copy the folder `figures_for_SI/` next to the SI source, since the snippets use these file names. Fig. S7: error vs. bending; Fig. S8: secondary-objective ablation (top: model-based, from the repository; bottom: data-driven, new); Fig. S9: coverage (repository figures, shown in Section 3.8).





Data-driven controller: IK network trained with / without the secondary term (20 control targets, offline)



2 Comparison with prior work (Comment 2)

Reviewer, Comment 2

The individual sensing and control elements of this paper have appeared in many papers. [. . .] The authors should explain what each close prior study established and identify the specific capability added here. The claim that no prior framework combines comparable sensing with closed-loop null-space control is too broad without this analysis.

Step 21 SI, end of the document: add Table S10 (comparison).

LaTeX to paste • `snippets/tab_S10_comparison.tex`

```
\begin{table*}[h]
\centering
\footnotesize
\caption{Comparison with closely related systems.}
\label{tab:S10_comparison}
\begin{tabular}{*{7}{>{\raggedright\arraybackslash}p{2.1cm}}}
\toprule
Work & Sensing & Robot & Estimated & Closed loop on the estimate & Redundancy resolution & Accuracy, rate \\
\midrule
Hofer et al.~\cite{hoferVisionBasedSensingApproach2021} & One internal camera per bellow (3), painted interior
pattern, LED illumination & Pneumatic fabric spherical arm, 2 rotational DoF & Orientation (2 angles) & Yes,
cascaded PI on the camera estimate & Actuation redundancy not exploited & $\approx 1^\circ$, 30~Hz \\
Thuruthel et al.~\cite{thuruthelLearningClosedLoop2017} & End-effector position feedback & Tendon-driven
continuum manipulator, 6 DoF & Tip position & Yes, learned inverse kinematics with feedback; with and without
external loads & Implicit in the learned inverse kinematics & -- \\
Han et al.~\cite{hanAnchoringMorphologicalRepresentations2025} & One internal camera tracking an anchor marker &
Soft metastructure modules & Anchor motion, force, full-body shape & Not demonstrated & -- & -- \\
Lu et al.~\cite{luAdaptiveOnlineLearning2024} & Fiber Bragg grating & Robotic colonoscope; multi-section
extensible soft manipulators & 3D shape & Yes, adaptive online-learning shape servoing & -- & -- \\
Yoo et al.~\cite{REF15} & Internal camera; model trained in simulation & Pneumatic soft actuator (100~mm) & 3D
shape & No & -- & 10.12~mm tip error \\
This work & One internal camera per section, no internal markers or patterns & Tendon-driven Trimmed Helicoid
arm, 1 and 3 sections & Section-tip poses (6 DoF), PPC curvature & Yes, model-based (PCC Jacobian) and
data-driven (learned inverse kinematics) & Null-space objective; secondary term in IK training & 3.99~mm,
$3.11^\circ$ (1 section); $\approx 8$~Hz \\
\bottomrule
\end{tabular}
\end{table*}
```

Preview (how it compiles)

Comparison with closely related systems.

Work	Sensing	Robot	Estimated	Closed loop on the estimate	Redundancy resolution	Accuracy, rate
Hofer et al. [ref]	One internal camera per bellow (3), painted interior pattern, LED illumination	Pneumatic fabric spherical arm, 2 rotational DoF	Orientation (2 angles)	Yes, cascaded PI on the camera estimate	Actuation redundancy not exploited	$\approx 1°$, 30 Hz
Thuruthel et al. [ref]	End-effector position feedback	Tendon-driven continuum manipulator, 6 DoF	Tip position	Yes, learned inverse kinematics with feedback; with and without external loads	Implicit in the learned inverse kinematics	–
Han et al. [ref]	One internal camera tracking an anchor marker	Soft metastructure modules	Anchor motion, force, full-body shape	Not demonstrated	–	–
Lu et al. [ref]	Fiber Bragg grating	Robotic colonoscope; multi-section extensible soft manipulators	3D shape	Yes, adaptive online-learning shape servoing	–	–
Yoo et al. [ref]	Internal camera; model trained in simulation	Pneumatic soft actuator (100 mm)	3D shape	No	–	10.12 mm tip error
This work	One internal camera per section, no internal markers or patterns	Tendon-driven Trimmed Helicoid arm, 1 and 3 sections	Section-tip poses (6 DoF), PPC curvature	Yes, model-based (PCC Jacobian) and data-driven (learned inverse kinematics)	Null-space objective; secondary term in IK training	3.99 mm, 3.11° (1 section); ≈ 8 Hz

Check before submission

How each row was checked:

- Hofer et al.: against the full text.
- Han et al.: against the project page.
- Yoo et al. (10.12 mm tip error, 100 mm actuator) and Lu et al.: against the abstracts.

Not confirmed from the full text:

- Thuruthel et al.: the sensor type (so the cell says only "end-effector position feedback") and the "implicit" redundancy.
- Han et al.: that a closed loop was "not demonstrated".
- The "–" cells of Lu et al.

Also replace `\cite{REF15}` with your key for Ref. [15] (Yoo et al.), and add the four new entries (Step 25) to the SI bibliography.

Step 22 Manuscript, Introduction: replace the novelty sentence by the comparison paragraph.

Find: "To the best of our knowledge, no framework has yet combined such sensing with closed-loop null-space control of continuum robots, either through model-based or learning-based approaches."

LaTeX to paste • `snippets/intro_comparison.tex`

```
Several systems combine embedded sensing with closed-loop control of soft or continuum arms. Hofer et al.\ placed
one camera inside each of the three fabric bellows of a pneumatic spherical arm, observed a painted interior
pattern under internal illumination, and regulated the two joint angles in closed loop with about $1^\circ$
accuracy at 30~Hz; the actuation redundancy of the arm was not exploited
\cite{hoferVisionBasedSensingApproach2021}. Thuruthel et al.\ learned closed-loop inverse kinematics with
end-effector feedback for a redundant tendon-driven continuum manipulator, with and without external loads
\cite{thuruthelLearningClosedLoop2017}. Han et al.\ tracked an internal anchor marker with a single onboard
camera to infer the full-body deformation and contact force of soft modules
\cite{hanAnchoringMorphologicalRepresentations2025}, and Yoo et al.\ estimated the 3D shape of a pneumatic
actuator from an internal camera with a model trained in simulation \cite{REF15}; neither closed a control loop
on the estimate. Lu et al.\ servoed the 3D shape of continuum and soft robots with adaptive online learning,
using fiber Bragg grating sensing \cite{luAdaptiveOnlineLearning2024}. The present work differs in two respects:
it distributes one internal camera per section along a multi-section tendon-driven arm, without internal markers,
patterns, or dedicated illumination, and it closes the loop on the camera-estimated pose with controllers that
resolve the actuation redundancy explicitly, either in the null space of a model-based Jacobian or through a
secondary objective in the training of a learned inverse kinematics. To the best of our knowledge, this
combination has not been reported.
```

Step 23 Manuscript, Introduction: replace the three sentences before it (scalability, effort).

Find: "Addressing the sensing and control challenges of continuum manipulators, this paper proposes a strategy [. . .] that scales

with the number of segments. Information from miniature cameras [...] full-body reconstruction. The reconstructed state is then used as feedback in both model-based and data-driven controllers, enabling tip regulation while reducing actuation effort."

LaTeX to paste • snippets/intro_scale_sentence.tex

```
Addressing the sensing and control challenges of continuum manipulators, this paper proposes a strategy for
vision-based reconstruction and control of soft robotic arms that can be extended to a different number of
sections. Information from miniature cameras embedded in the sections is fused within a single CNN to achieve
full-body reconstruction. The estimated tip pose is then used as feedback in both model-based and data-driven
controllers, enabling tip regulation while keeping the tendon lengths centered in their actuation range.
```

Step 24 Manuscript, Introduction: replace the list of contributions (items 1–4).

This also covers comments 1 (role of the encoders), 3 (scalability) and 11 (markerless).

LaTeX to paste • snippets/intro_contributions.tex

```
\begin{enumerate}
\item A distributed vision-based sensing strategy for soft arms that uses miniature cameras embedded in each
section of the robot. It provides dense feedback that is markerless at inference (external motion-capture markers
are used only to generate training labels) for tip-pose estimation and full-body curvature reconstruction, and
can be extended to arms with a different number of sections through modular hardware and a lightweight CNN whose
backbone is transferred and retrained; we validated arms with one and three sections.
\item Integration of embedded vision with model-based null-space control. We close the loop on the
camera-estimated pose with a Jacobian-based Piecewise Constant Curvature velocity controller, in which the motor
encoders provide the tendon lengths for the Jacobian and the null-space term.
\item A data-driven framework for both sensing and control. We train learning-based inverse kinematics models
that generate actuation commands from the camera-estimated pose without encoder feedback, with a secondary
objective in the training loss.
\item Experimental validation of pose regulation with both controllers on single- and multi-section arms,
including a constant tip payload, and an offline evaluation of the secondary objectives.
\end{enumerate}
```

Step 25 Bibliography: add the four entries.

LaTeX to paste • snippets/bib_new.tex

```
@article{hoferVisionBasedSensingApproach2021,
  title = {A Vision-Based Sensing Approach for a Spherical Soft Robotic Arm},
  author = {Hofer, Matthias and Sferrazza, Carmelo and D'Andrea, Raffaello},
  journal = {Frontiers in Robotics and AI},
  volume = {8},
  pages = {630935},
  year = {2021},
  doi = {10.3389/frobt.2021.630935}
}

@article{thuruthelLearningClosedLoop2017,
  title = {Learning Closed Loop Kinematic Controllers for Continuum Manipulators in Unstructured Environments},
  author = {Thuruthel, Thomas George and Falotico, Egidio and Manti, Mariangela and Pratesi, Andrea and
  Cianchetti, Matteo and Laschi, Cecilia},
  journal = {Soft Robotics},
  volume = {4},
  number = {3},
  pages = {285--296},
  year = {2017},
  doi = {10.1089/soro.2016.0051}
}

@article{hanAnchoringMorphologicalRepresentations2025,
  title = {Anchoring Morphological Representations Unlocks Latent Proprioception in Soft Robots},
  author = {Han, Xudong and Guo, Ning and Xu, Ruoyu and Wan, Fang and Song, Chaoyang},
  journal = {Advanced Intelligent Systems},
  volume = {7},
  number = {12},
  pages = {e202500444},
  year = {2025},
  doi = {10.1002/aisy.202500444}
}

@article{luAdaptiveOnlineLearning2024,
  title = {Adaptive Online Learning and Robust 3-D Shape Servoing of Continuum and Soft Robots in Unstructured
  Environments},
  author = {Lu, Yiang and Chen, Wei and Lu, Bo and Zhou, Jianshu and Chen, Zhi and Dou, Qi and Liu, Yun-Hui},
  journal = {Soft Robotics},
  volume = {11},
  number = {2},
```

```
  pages = {320--337},
  year = {2024},
  doi = {10.1089/soro.2022.0158}
}
```

Response letter text (LaTeX) • `snippets/resp_2.tex`

```
We thank the reviewer for these references. We added Hofer et al., Thuruthel et al., Han et al., and Lu et al.\
(Yoo et al.\ was already cited as Ref.~[15]) and a paragraph in the Introduction that states what each study
established, together with a comparison table in the SI (Table~S10). Hofer et al.\ obtain a higher orientation
accuracy (about $1^\circ$ at 30~Hz) than our single-section estimate ($3.11^\circ$ at about 8~Hz), on a two-angle
spherical joint with an internal painted pattern and illumination; our system estimates the full 6-DoF tip pose
of an extensible section without internal markers and is applied to three sections. Thuruthel et al.\ learned
closed-loop inverse kinematics with end-effector feedback, Han et al.\ and Yoo et al.\ estimate shape from
internal cameras without closing a control loop, and Lu et al.\ servo the shape with fiber Bragg grating sensing.
The capability we add is the combination of distributed internal cameras on a multi-section arm with closed-loop
control that explicitly exploits the actuation redundancy, through model-based null-space control and through a
learned inverse kinematics with a secondary objective. We narrowed the novelty statement in the Introduction and
the contribution list to this combination.
```

3 Everything else, by topic

3.1 Editor's requirements

Editor

Title succinct, descriptive and accurate [. . .] define all acronyms when first mentioned (the title, abstract, and Table of Contents text must each be understandable as standalone texts). Check that the text and axis labels in all graphics are correct [. . .] all parts, insets, and footnotes of Figures [. . .] are mentioned in the caption. [. . .] copyright permission [. . .] full article titles, and optionally, DOIs.

Step 26 Acronyms: define at first use, in the main text and again in the SI.

- Text: PCC, PPC, CNN, MLP, GAP, IK, FK, TPU, PLA, FBG, IMU, RRT, SLERP, DoF, SE(3)/SO(3), Grad-CAM, ELU, BN.
- Figures (defined by the new captions): BW, FE, FC (Fig. 1); CL, OL (Figs. 5 and 8).
- The abstract and title contain no acronyms. If you have a ToC text, check it too.

Step 27 References: add a DOI to every entry that has one.

Step 28 Response letter.

Response letter text (LaTeX) • snippets/resp_G.tex

```
\textbf{Title.} We kept the title, which states the method (internal-camera reconstruction and closed-loop
control) and the system (soft robotic arms).

\textbf{Suggested references.} We added the references suggested by the reviewer (Comment~2).

\textbf{Acronyms.} We proofread the manuscript and define every acronym at its first use in the main text and in
the SI, including those that appear only in figures (e.g., BW, FE, and FC in Fig.~1, and CL and OL in Figs.~5 and
8). The title and abstract contain no undefined acronyms.

\textbf{Figures and captions.} We checked the text and axis labels of all figures and revised the captions so
that every panel and inset is described (Comment~10).

\textbf{Copyright.} All figures were created by the authors for this manuscript; no reprinted or adapted
material, logos, or trademark images are included.

\textbf{References.} The reference list gives full article titles, and we added DOIs where available.
```

3.2 Contribution, scope and claims (Comments 3, 11, 12, 27)

Reviewer, Comment 3

[. . .] the present method is not plug-and-play across arbitrary arm lengths and is not validated on an unseen section count. The statement that only the first layer must be adjusted also conflicts with Fig. 3 because the regression head changes as well.

Step 29 Manuscript, Section 3.1, second paragraph: replace three sentences.

This also aligns the text with the two-layer heads of the multi-section networks.

Find: "To process the image batches, the first 2D kernel is modified and preceded by a normalization layer. This is the only non-pre-trained component of the feature extractor, making the sensing design scalable, as adapting to a different number of sections requires adjusting only the first layer (Fig. 3a). At the end of the backbone, we apply Global Average Pooling (GAP) followed by a fully connected head that maps directly to the regression outputs."

LaTeX to paste • snippets/sec31_scalability.tex

```
To process the image batches, the first convolutional layer is adapted to the number of stacked images and
preceded by a normalization layer. Adapting the network to a different number of sections requires changing this
input layer and the regression head ($6n$ pose outputs and $n$ sets of curvature parameters) and retraining, with
the remaining backbone transferred (Section~3.1.1); we validated this for $n = 1$ and $n = 3$. At the end of the
backbone, we apply Global Average Pooling (GAP) followed by a regression head: one fully connected layer for the
single-section networks and two for the multi-section networks (Fig.~3a).
```

Step 30 Manuscript, Section 3.1, third paragraph: replace (architectures of the kinematic networks, as in the code).

Find: "Forward and inverse kinematics are modeled using a custom MLP architecture, detailed in Fig. 3b. The network utilizes residual blocks to prevent vanishing gradients, yielding the inverse mappings [. . .], and the forward mapping [. . .]."

LaTeX to paste • `snippets/sec31_mlp.tex`

```
The single-section inverse kinematics $N_l^1: \mathbf{t} \mapsto \mathbf{l}$ uses the residual MLP of Fig.~3b,
whose residual blocks mitigate vanishing gradients; the same architecture is used for $N_C$. The multi-section
forward and inverse mappings $N_p^3: \mathbf{l} \mapsto \mathbf{p}$ and $N_l^3: \mathbf{p} \mapsto \mathbf{l}$
are MLPs with four hidden layers of 128 units and ELU activations (Table~S11).
```

Step 31 Manuscript, Section 5, first paragraph: replace.

Find: "This work proposes a scalable framework for vision-based proprioception and control of soft continuum robotic arms. [. . .] secondary tasks within the null space."

LaTeX to paste • `snippets/concl_first.tex`

```
This work proposes a framework for vision-based proprioception and control of soft continuum robotic arms.
Miniature cameras embedded in each section feed deep learning algorithms that estimate the section-tip poses and
reconstruct the full shape of the robot. The estimated poses close the loop of two control strategies,
model-based and data-driven, which explicitly exploit kinematic redundancy to accomplish secondary tasks within
the null space; the shape reconstruction is evaluated as a separate sensing output.
```

The Introduction and contribution 1 are handled in Step 23 and Step 24.

Response letter text (LaTeX) • `snippets/resp_3.tex`

```
We agree. The input layer and the regression head both depend on the number of sections, and separate networks
were trained for the one- and three-section arms. We corrected the statement in Section~3.1 and rephrased the
Introduction, the contribution list, and the Conclusion accordingly: the backbone is transferred and the network
is retrained, and only $n = 1$ and $n = 3$ were validated. We also removed the word ``scalable'' where it implied
that no retraining is needed.
```

Reviewer, minor comments 11 and 12

Use 'markerless at measured inferences' rather than simply 'markerless.' [. . .] Use either 'section' or 'segment' consistently throughout the manuscript, figures, algorithms, and SI.

Step 32 "Markerless": its only occurrence is contribution 1, already replaced (Step 24).

Step 33 Replace "segment" everywhere. Search your sources for "egment" (case-insensitive); the expected hits are:

Where	Current	New
Intro, "Addressing. . . "	that scales with the number of segments	handled in Step 23
Sec. 2.1, 1st paragraph	corresponds to the first segment of the multi-section arm	. . . the first section . . .
Fig. 1 caption	three independent segments	handled in Step 1
Sec. 2.2	each segment bends in any direction	each section bends . . .
Sec. 2.2, PPC paragraph	central axis of a segment	handled in Step 52
Sec. 2.2, markers	placed along each segment	handled in Step 55
Sec. 3.2, 2nd paragraph	Curvature parameters C_j for segment j	. . . for section j
Sec. 3.2, 2nd paragraph	for the multi-segment case (twice)	for the multi-section case
Sec. 3.2, 2nd paragraph	segment curvature parameters	section curvature parameters
Fig. 5 caption	single-segment arm	handled in Step 8
Heading 4.1	Single-Segment Reconstruction and Control	Single-Section Reconstruction and Control
Fig. 8 caption	multi-segment arm	handled in Step 15
Heading 4.2	Multi-Segment Reconstruction and Control	Multi-Section Reconstruction and Control
Sec. 4.2.1	at the interfaces between segments	. . . between sections
Sec. 4.2.1	consistent with the single-segment case	handled in Step 87
Sec. 5	cameras embedded in each segment	handled in Step 31
SI 2.1	each arc included in a segment	. . . in a section
SI 2.1	arc lengths of the cylinder soft segment	. . . of the soft section
SI Fig. S1 caption	multi-segment robot	handled in Step 16
SI 3.2	the different segments	the different sections
SI 4.1	single- and multi-segment manipulators; multi-segment robot	handled in Step 66
SI 4.2	multi-segment manipulator	multi-section manipulator
SI Table S1	outer boundary segment	handled in Step 59
SI 4.3	across segments (twice)	across sections

Where	Current	New
SI 4.3	from a source segment to a target segment	from a source section to a target section
SI 6.1	with respect to the preceding segment	. . . the preceding section
Figure files	any “segment” inside graphics	check Figs. 1, 5, 8, S1

Response letter text (LaTeX) • snippets/resp_11.tex

```
We now write ``markerless at inference'' and explain at first use that external motion-capture markers are used
only to generate the training labels.
```

Response letter text (LaTeX) • snippets/resp_12.tex

```
We now use ``section'' throughout the manuscript, figures, algorithms, and SI.
```

Reviewer, minor comment 27 and Comment 9 (last sentence)

The discussion should distinguish demonstrated capabilities from proposed applications. Obstacle avoidance, whole-body grasping, surgical guidance, and force or contact sensing are future directions and were not evaluated here. [. . .] Claims of robustness to lighting or hardware variation should be limited unless such variation was explicitly tested.

Step 34 Manuscript, Section 5, second paragraph: replace the last two sentences.

Find: “Experimental results demonstrate that our sensing approach reaches competitive performance in reconstructing both tip pose and full-body curvature, with preserved accuracy at workspace boundaries. Closed-loop control achieves near-sensing performance and outperforms open-loop alternatives, while being robust to external disturbances.”

LaTeX to paste • snippets/concl_second.tex

```
Experimental results demonstrate that our sensing approach reaches competitive performance in reconstructing both
tip pose and full-body curvature, with accuracy that does not degrade at large bending angles for the
single-section arm. Closed-loop control achieves near-sensing performance, outperforms open-loop alternatives,
and compensates a constant 50~g tip payload.
```

Step 35 Manuscript, Section 5, third paragraph: replace the whole paragraph.

Find: “This sensing and control framework endows soft robots with accurate motion capabilities. [. . .] to actively shape the arm for obstacle avoidance while maintaining the desired tip pose.”

LaTeX to paste • snippets/concl_third.tex

```
Potential applications, not evaluated in this work, include search-and-rescue or inspection within confined and
cluttered environments and minimally invasive surgery. In such settings, full-body curvature feedback from
embedded cameras could be used by the null-space controller to shape the arm for obstacle avoidance while
maintaining the desired tip pose.
```

Step 36 Manuscript, Section 5, limitations paragraph: replace the lighting sentence.

Find: “Illumination passes through the opaque covering, and variability is mitigated through image augmentations during training.”

LaTeX to paste • snippets/concl_lighting.tex

```
Illumination passes through the opaque covering; image augmentations were applied during training to account for
residual brightness variations, but robustness to changes in lighting or hardware was not tested.
```

Step 37 Manuscript, Section 5, same paragraph: change “suitable for applications such as inspection or surgical guidance” to “which may suit applications such as inspection”.

Step 38 Manuscript, Section 5: replace the effort sentence.

Find: “In this paper, redundancy was exploited to balance tendon effort, but the same methodology applies to shaping arm configurations for obstacle avoidance while tracking a desired end-effector trajectory.”

LaTeX to paste • snippets/concl_effort.tex

```
In this paper, redundancy was exploited to keep the tendon lengths centered in their actuation range, but the
same methodology applies to shaping arm configurations for obstacle avoidance while tracking a desired
end-effector trajectory.
```

Step 39 Manuscript, Section 5: insert after “. . . for closed-loop control in dynamic conditions.”

LaTeX to paste • snippets/concl_limits_add.tex

```
The control evaluation used 20 targets, each visited once, and planar circular trajectories; repeated trials,
non-planar and workspace-boundary trajectories, and further load magnitudes and directions remain to be tested.
The training, validation, and test sets were drawn from the same acquisition sessions, so session-blocked
evaluation is also left to future work. The secondary objectives were evaluated offline through the kinematic
models (SI Section~5.3) rather than in hardware experiments.
```

Response letter text (LaTeX) • snippets/resp_27.tex

```
We revised Section~5 to separate the capabilities evaluated in this work from future applications. Obstacle
avoidance, whole-body grasping, surgical and inspection applications, and force or contact sensing are now
presented explicitly as future directions that were not evaluated here.
```

3.3 How sensing enters the control loop (Comments 1, 18, 19, 20)

Reviewer, Comment 1

[. . .] both controllers use feedback from the pose network, while the curvature network is evaluated separately and does not enter either control law. Algorithm 1 also uses a PCC inverse calculation to initialize tendon lengths from section-tip positions. [. . .] the resulting estimate appears to be replaced by the motor-encoder measurement at the first control iteration, before it affects the control command. The authors should explain this initialization or remove it. [. . .]

Algorithm 1 Model-based feedback controller

Set target p_{goal} and convergence threshold γ
Measure current pose p from $(f_p^3(I))$
Measure tip positions $[t_I, t_{II}, t_{III}]$ from $(f_p^3(I))$
Estimate $l \leftarrow l(t_I, t_{II}, t_{III})$ via PCC
while $d_{SE(3)}(p, p_{goal}) > \gamma$ do
  Compute next reference $p_{ref} \leftarrow (\mathcal{W}^3(p, p_{goal}))_1$
  Measure l from motor encoders
  Compute Jacobian $J \leftarrow J_{l2p}(l)$
  Compute damped pseudoinverse $J^{\#} \leftarrow J^{\intercal}(JJ^{\intercal} + \lambda^2 I)^{-1}$
  Compute null-space projector $N \leftarrow I - J^{\#}J$
  Move motors by $\delta l \leftarrow J^{\#} \cdot (p_{ref} - p) + N \cdot (\bar{l} - l)$
  Measure p from $(f_p^3(I))$
end while

What the code does (`MULTI_controller.py`, constructor and control loop; the same in `SINGLE_controller.py`):

1. The PCC estimate from the camera tip positions (marker ①) gives $\mathbf{l}_0$.
2. The encoder offset is set once, $\mathbf{l}_{\text{off}} = \mathbf{l}_{\text{enc}} - \mathbf{l}_0$ (`DELTAL_reference`).
3. Inside the loop (marker ②), the controller uses $\mathbf{l} = \mathbf{l}_{\text{enc}} - \mathbf{l}_{\text{off}}$.

So the line is used: it puts the encoder readings in the PCC tendon frame. Keep it and say so.

Step 40 Manuscript, Abstract: replace the third sentence.

Find: "Miniature cameras embedded in each section provide dense proprioceptive feedback, while convolutional neural networks map multi-view images to both tip pose and full-body reconstruction using a polynomial curvature model."

LaTeX to paste • snippets/abstract_sentence2.tex

```
Miniature cameras embedded in each section provide dense proprioceptive feedback, while convolutional neural
networks map multi-view images to both tip pose and full-body reconstruction using a polynomial curvature model.
The estimated tip poses close the control loop, while the curvature reconstruction is evaluated as a separate
sensing output.
```

Step 41 Manuscript, Abstract: replace the last sentence.

Find: "Experiments demonstrate that embedded vision enables accurate full-body state reconstruction, reliable redundancy resolution, and robust regulation even under external disturbances."

LaTeX to paste • snippets/abstract_last.tex

```
Experiments show that embedded vision enables accurate tip-pose estimation and model-constrained full-body
reconstruction, and closed-loop pose regulation with redundancy resolution that compensates a constant tip
payload.
```

Step 42 Manuscript, Section 3.3: insert both snippets after "... until the distance to the goal falls below a given threshold γ."

LaTeX to paste • snippets/sec33_feedback.tex

```
In both controllers, the feedback signal is the tip pose estimated by the pose CNN from the camera images. The
model-based controllers additionally read the tendon lengths from the motor encoders to evaluate the PCC Jacobian
and the null-space term, whereas the data-driven controllers do not use encoders. The curvature CNNs are not part
of either control law; full-body reconstruction is evaluated as a separate sensing output (Sections~4.1.2 and
4.2.2).
```

LaTeX to paste • snippets/notation_rule.tex

```
Throughout, $\mathbf{J}_{a \to b} = \partial b / \partial a$ denotes the Jacobian of the mapping from $a$ to $b$;
analytical (PCC) Jacobians are forward Jacobians ($\mathbf{J}_{l \to t}$, $\mathbf{J}_{l \to p}$) and
$\mathbf{J}^{\#}$ is their damped pseudoinverse, while hats denote Jacobians of the learned inverse-kinematics
networks ($\hat{\mathbf{J}}_{t \to l}$, $\hat{\mathbf{J}}_{p \to l}$).
```

Step 43 **Manuscript, Algorithm 1: replace the whole algorithm (algpseudocode syntax; adapt if you use another package).**

LaTeX to paste • snippets/alg1.tex

```
\begin{algorithm}[t]
\caption{Model-based feedback controller}
\begin{algorithmic}
\State Set target $\mathbf{p}_{\mathrm{targ}}$ and convergence threshold $\gamma$
\State Measure current pose $\mathbf{p}$ from $f_p^3(\mathbf{I})$
\State Measure tip positions $[\mathbf{t}_{\mathrm{I}}, \mathbf{t}_{\mathrm{II}}, \mathbf{t}_{\mathrm{III}}]$
from $f_p^3(\mathbf{I})$
\State Estimate $\mathbf{l}_0 \gets \mathbf{l}(\mathbf{t}_{\mathrm{I}}, \mathbf{t}_{\mathrm{II}},
\mathbf{t}_{\mathrm{III}})$ via PCC
\State Set encoder offset $\mathbf{l}_{\mathrm{off}} \gets \mathbf{l}_{\mathrm{enc}} - \mathbf{l}_0$
\While{$d_{SE(3)}(\mathbf{p}, \mathbf{p}_{\mathrm{targ}}) > \gamma$}
\State Compute next reference $\mathbf{p}_{\mathrm{ref}} \gets W^3(\mathbf{p}, \mathbf{p}_{\mathrm{targ}})$
\State Measure $\mathbf{l} \gets \mathbf{l}_{\mathrm{enc}} - \mathbf{l}_{\mathrm{off}}$ from motor encoders
\State Compute Jacobian $\mathbf{J} \gets \mathbf{J}_{l \to p}(\mathbf{l})$
\State Compute damped pseudoinverse $\mathbf{J}^{\#} \gets \mathbf{J}^T (\mathbf{J}\mathbf{J}^T + \lambda^2
\mathbf{I})^{-1}$
\State Compute null-space projector $\mathbf{N} \gets \mathbf{I} - \mathbf{J}^{\#}\mathbf{J}$
\State Move motors by $\delta\mathbf{l} \gets \mathbf{J}^{\#}(\mathbf{p}_{\mathrm{ref}} - \mathbf{p}) +
\mathbf{N}(\bar{\mathbf{l}} - \mathbf{l})$
\State Measure $\mathbf{p}$ from $f_p^3(\mathbf{I})$
\EndWhile
\end{algorithmic}
\end{algorithm}
```

Step 44 **Manuscript, Section 3.3.2: replace the sentence “The model-based controller is summarized in Alg. 1.” (the snippet starts with it).**

LaTeX to paste • snippets/sec332_offset.tex

```
The model-based controller is summarized in Alg.~1. Before the loop, the tendon lengths $\mathbf{l}_0$
corresponding to the camera-estimated section-tip positions are computed through the inverse PCC map (Eq.~2) and
used once to set the encoder offset $\mathbf{l}_{\mathrm{off}} = \mathbf{l}_{\mathrm{enc}} - \mathbf{l}_0$.
Inside the loop, the encoder readings are expressed in the same PCC tendon-length frame, $\mathbf{l} =
\mathbf{l}_{\mathrm{enc}} - \mathbf{l}_{\mathrm{off}}$, which is used by the Jacobian and by the null-space term.
The single-section controller and the data acquisition use the same offset.
```

Response letter text (LaTeX) • snippets/resp_1.tex

```
We thank the reviewer for this careful reading. The reviewer is correct that the curvature reconstruction does
not enter either control law: both controllers close the loop on the tip pose estimated from the camera images,
the model-based controller additionally uses the motor encoders to evaluate the PCC Jacobian and the null-space
term, and the curvature estimate is evaluated as a separate sensing capability. We now state this explicitly in
the Abstract, the contribution list, Section~3.3, the caption of Fig.~4c (which now shows the encoder path), and
the Conclusion.

The initial PCC estimate in Algorithm~1 is not discarded. The tendon lengths $\mathbf{l}_0$ obtained from the
camera-estimated section-tip positions through the inverse PCC map are used once, before the loop, to set the
encoder offset $\mathbf{l}_{\mathrm{off}} = \mathbf{l}_{\mathrm{enc}} - \mathbf{l}_0$, so that the encoder
readings inside the loop are expressed in the PCC tendon-length frame used by $\mathbf{J}(\mathbf{l})$ and by the
null-space term $\mathbf{N}(\bar{\mathbf{l}} - \mathbf{l})$. Algorithm~1 and Section~3.3.2 now show this step
explicitly.
```

Response letter text (LaTeX) • `snippets/resp_18.tex`

```
Please see our response to Comment~1: the PCC estimate sets the encoder offset used inside the loop, and
Algorithm~1 now shows this step explicitly.
```

Reviewer, minor comments 19 and 20

Use consistent Jacobian notation. J_t2l, J_l2p, and their learned representations currently alternate between a forward Jacobian, an inverse Jacobian, and a derivative of an inverse network. [. . .] State whether the open-loop baselines use the same measured initial pose, planner, gains, stopping limits, and commanded waypoint sequence as the closed-loop methods. 'Open-loop' and 'feedforward' should be used consistently.

The rule is defined by `notation_rule` (Step 42): $\mathbf{J}_{a\to b} = \partial b/\partial a$. PCC Jacobians are forward ($\mathbf{J}_{l\to t}$, $\mathbf{J}_{l\to p}$); learned ones have a hat ($\hat{\mathbf{J}}_{t\to l}$, $\hat{\mathbf{J}}_{p\to l}$). At present, Section 3.3.1 calls the forward PCC Jacobian "J_{t2l}".

Step 45 Manuscript, Section 3.3.1, first sentence of page 7: replace.

Find: "with encoder position feedback. Once l is measured from the motor encoders, the forward derivative J is computed as J_t2l(l)."

LaTeX to paste • `snippets/sec331_jacobian.tex`

```
with encoder position feedback. Once $\mathbf{l}$ is measured from the motor encoders, the forward Jacobian
$\mathbf{J}_{l \to t}(\mathbf{l}) = \partial \mathbf{t} / \partial \mathbf{l}$ is computed from the PCC model.
```

Step 46 Manuscript, Eq. 7: replace.

LaTeX to paste • `snippets/eq7.tex`

```
\begin{equation}
\hat{\mathbf{J}}_{t \to l} = \left. \frac{\partial N_l^1}{\partial \mathbf{t}} \right|_{\mathbf{t}}
\end{equation}
```

Step 47 Manuscript, Algorithm 2: replace the two lines "Compute J_{p2l}. . . " and "Move motors by. . . ".

LaTeX to paste • `snippets/alg2_lines.tex`

```
\State Compute $\hat{\mathbf{J}}_{p \to l} \gets \left. \partial N_l^3 / \partial \mathbf{p}
\right|_{\mathbf{p}}$
\State Move motors by $\delta\mathbf{l} \gets \hat{\mathbf{J}}_{p \to l} (\mathbf{p}_{\mathrm{ref}} -
\mathbf{p})$
```

Step 48 SI, Table S2: replace the two Jacobian rows.

LaTeX to paste • `snippets/tabS2_jac_rows.tex`

```
$\mathbf{J}_{l \to t}, \mathbf{J}_{l \to p}, \mathbf{J}^{\#}$ & Analytical (PCC) forward Jacobians, $\partial
\mathbf{t}/\partial \mathbf{l}$ and $\partial \mathbf{p}/\partial \mathbf{l}$, and their regularized
Moore-Penrose pseudoinverse. \\
$\hat{\mathbf{J}}_{t \to l}, \hat{\mathbf{J}}_{p \to l}$ & Jacobians of the learned inverse kinematics, $\partial
N_l^1/\partial \mathbf{t}$ and $\partial N_l^3/\partial \mathbf{p}$. \\
```

Step 49 SI, Fig. S3: symbols as in Step 18.

Step 50 Manuscript, Section 3.3.1: replace the open-loop sentence.

Find: "The same algorithms are also employed in open-loop control as a baseline, where robot states are internally updated through either model-based or data-driven forward kinematics."

LaTeX to paste • `snippets/sec331_openloop.tex`

```
The same algorithms are also employed in open-loop control as a baseline. The open-loop controllers use the same
planner, waypoint spacing $\varepsilon$, damping $\lambda$, and target sequence as the closed-loop ones. Their
initial state is measured once by the motion capture system and then propagated internally through the
model-based or data-driven kinematics instead of the camera estimate, and the full waypoint sequence is executed
without the convergence check.
```

Step 51 Replace "feedforward" by "open-loop" (keep the occurrence in the Introduction, which is about iterative learning control).

- Section 4.1.4: "the data-driven feedforward controller" → "the data-driven open-loop controller".
- Section 4.2.5: "feedforward accuracy is lower" → "open-loop accuracy is lower".

- Captions of Figs. 6, 9 and 10: done in Part 1, including "feedback controller" → "closed-loop controller".

Response letter text (LaTeX) • `snippets/resp_19.tex`

```
We now use a single notation, defined in Section~3.3: $\mathbf{J}_{a \to b} = \partial b / \partial a$. The
analytical PCC Jacobians are forward Jacobians ($\mathbf{J}_{l \to t}$, $\mathbf{J}_{l \to p}$) with damped
pseudoinverse $\mathbf{J}^{\#}$, and the Jacobians of the learned inverse kinematics are denoted with a hat
($\hat{\mathbf{J}}_{t \to l}$, $\hat{\mathbf{J}}_{p \to l}$). The notation is used consistently in Eqs.~7--10,
Algorithms~1--2, Fig.~S3, and Table~S2.
```

Response letter text (LaTeX) • `snippets/resp_20.tex`

```
Section~3.3.1 now states that the open-loop baselines use the same planner, waypoint spacing, damping, and target
sequence as the closed-loop controllers. Their initial state is measured once by motion capture and then
propagated with the model-based or data-driven kinematics instead of the camera estimate, and they execute the
full waypoint sequence without a convergence check. We use ``open-loop'' throughout.
```

3.4 Kinematic model (Comment 5)

Reviewer, Comment 5

[. . .] The paper should consistently use either normalized arc length $u = x/L$ or dimensional arc length $x \in [0, L]$, and revise Eqs. 4–6, SI Eqs. S3–S9, the translation integral, nomenclature, and coefficient units accordingly.

The code uses the dimensional arc length everywhere: labels are evaluated at $S/3$, $2S/3$ and S in mm, and the reconstruction integrates over $[0, S]$. SI Eqs. S5–S9 do the same. So the main text changes and Eqs. S5–S9 stay. The SI sentence before Eq. S5 also uses "$s = 1/3, 2/3, 1$" and must change (Step 57).

Step 52 Manuscript, Section 2.2, PPC paragraph and Eq. 4: replace.

Find: "However, soft arms undergo deformations due to external forces and self-weight, [. . .] polynomial in the normalized arc length $s \in [0, 1]$, with the local bending angle $\theta(s) = \int_0^s \kappa(v) L\, dv$. For example, in the quadratic case: [Eq. 4]"

LaTeX to paste • `snippets/eq4_text.tex`

```
However, soft arms undergo deformations due to external forces and self-weight, requiring more complex shape
approximations for full-body proprioception. We adopt the Piecewise Polynomial Curvature (PPC) model [31], which
retains the $\boldsymbol{\xi}$-parameters. It expresses the curvature $\kappa(x)$ of the central axis of a
section as a polynomial in the arc length $x \in [0, L]$, with the local bending angle $\theta(x) = \int_0^x
\kappa(v)\, dv$. For example, in the quadratic case:
\begin{equation}
\theta(x) = c_0 x + \frac{1}{2} c_1 x^2 + \frac{1}{3} c_2 x^3, \qquad x \in [0, L]
\end{equation}
where $c_0$, $c_1$, and $c_2$ have units of mm$^{-1}$, mm$^{-2}$, and mm$^{-3}$.
```

Step 53 Manuscript, Eq. 5a: replace s by x (and $\theta(s)$ by $\theta(x)$). Eq. 5b: replace with the snippet.

LaTeX to paste • `snippets/eq5b.tex`

```
\mathbf{t}(x) = \left[ c_\phi \int_0^x s_{\theta(v)}\, dv, \;\; s_\phi \int_0^x s_{\theta(v)}\, dv, \;\; \int_0^x
c_{\theta(v)}\, dv \right]^T
```

Step 54 Manuscript, Section 2.2: insert after Eq. 5 (Comment 4, assumptions).

LaTeX to paste • `snippets/assumptions.tex`

```
Within each section, the PPC model assumes planar bending with a bending direction $\phi$ that is constant along
the arc length, and no torsion; the curvature varies along the section only through the polynomial coefficients.
```

Step 55 Manuscript, markers paragraph (just before Eq. 6): replace the first three sentences.

Find: "For capturing ground truth pose, reflective markers for motion capture are placed along each segment at intervals of 1/3 of its length. We refer to these bodies as subsections of the module. They provide ground-truth bending direction and angle estimates [. . .] at $s = 1/3, 2/3, 1$ (details in the Supporting Information)."

LaTeX to paste • `snippets/eq6_text.tex`

```
For capturing ground truth pose, reflective markers for motion capture are placed along each section at intervals
of $L/3$. We refer to these bodies as subsections of the module. They provide ground-truth bending direction and
angle estimates $\hat{\phi}_x$, $\hat{\theta}_x$ at $x = L/3, 2L/3, L$ (details in the Supporting Information).
```

Step 56 Manuscript, Eq. 6 and the sentence after it: replace.

LaTeX to paste • snippets/eq6.tex

```
\begin{equation}
\mathop{\arg\min}_{c_0, c_1, c_2 \in \mathbb{R}} \sum_{x \in \{L/3,\, 2L/3,\, L\}} \left\| \mathbf{R}_x(c_0, c_1, c_2) - \hat{\mathbf{R}}_x \right\|_F^2
\end{equation}
The bending direction $\phi$ is the mean of $\hat{\phi}_{L/3}$, $\hat{\phi}_{2L/3}$, $\hat{\phi}_{L}$, while $L$ is the average tendon length according to Eq.~2.
```

Step 57 **SI, Section 2.2, sentence before Section 2.2.1: replace.**

Find: "Polynomial coefficients are initialized analytically by substituting $s = 1/3, 2/3, 1$ into $\theta(s) = [\dots]$ and solving the corresponding linear systems according to the optimization algorithm described in [2]."

LaTeX to paste • snippets/si22_init.tex

```
Polynomial coefficients are initialized analytically by substituting $x = L/3, 2L/3, L$ into $\theta(x) = c_0 x + \frac{1}{2} c_1 x^2 + \frac{1}{3} c_2 x^3$ and solving the corresponding linear systems according to the optimization algorithm described in [2].
```

Step 58 **SI, Eqs. S3 and S4: replace the variable s by x. Eqs. S5–S9 stay.**

Step 59 **SI, Table S1: replace the rows of s, $\theta(s)$, $\kappa(s)$, c_0, c_1, c_2 and L_i.**

LaTeX to paste • snippets/tabS1_rows.tex

```
$x$ & Arc length along a section axis ($x \in [0, L]$). \\
$\theta(x)$ & Local bending angle along the central axis at arc length $x$. \\
$\kappa(x)$ & Polynomial curvature function at arc length $x$. \\
$c_0, c_1, c_2$ & Coefficients of the polynomial curvature approximation (units mm$^{-1}$, mm$^{-2}$, mm$^{-3}$). \\
$L_i$ & Length of the $i$-th outer boundary arc ($i \in \{1, 2, 3, 4\}$). \\
```

Response letter text (LaTeX) • snippets/resp_5.tex

```
We thank the reviewer. The implementation and the SI equations use the dimensional arc length $x \in [0, L]$. We corrected the main text accordingly (Eqs.~4--6, the translation integral without the factor $L$, and the evaluation points $x = L/3, 2L/3, L$), the initialization sentence in SI Section~2.2, and the nomenclature in Table~S1, which now gives the units of $c_0$, $c_1$, and $c_2$ (mm$^{-1}$, mm$^{-2}$, mm$^{-3}$).
```

3.5 Ground truth, datasets and split (Comments 4, 8; SI comment 31)

Reviewer, Comment 4 (end)

Wherever possible, reconstructed backbone points should be compared directly with motion-capture marker positions, and uncertainty in N_C should be propagated into the reported errors. Otherwise, the result should be described as model-constrained reconstruction, with its assumptions stated explicitly, including planar bending within each section, no torsion, and no arc-dependent bending direction.

As agreed, answer with the model-constrained framing and the explicit assumptions (Step 54).

Step 60 **Manuscript, Section 4.2.2: insert at the end of the section.**

LaTeX to paste • snippets/sec422_modelconstrained.tex

```
For the multi-section arm, inter-section marker occlusion prevented tracking the subsection markers during data collection, so the curvature labels were obtained from the motion-captured section-tip poses through the network $N_C$ (SI Section~4.2). The multi-section results therefore quantify a model-constrained reconstruction, under the assumptions of Section~2.2 (planar bending within each section, no torsion, and a bending direction that is constant along each section), rather than the agreement with directly measured backbone points.
```

Step 61 **Manuscript, Fig. 7 caption: done in Step 12 (it says "model-constrained").**

Step 62 **SI, Section 4.2: insert after "...treating the robot as a modular extension of the single-section unit."**

LaTeX to paste • snippets/si42_nc.tex

```
$N_C$ was trained on the single-section dataset, mapping the tip pose to the curvature labels of Section~2.2,
with the same split as the other networks (Table~S11). On the held-out single-section test set ($N = 500$), its
3D error at $L/3$, $2L/3$, and $L$ is 0.38, 1.20, and 2.49~mm for the constant model, 0.42, 1.35, and 2.76~mm for
the affine model, and 0.43, 1.37, and 2.94~mm for the quadratic model. Fig.~S2b reports its error on 20
configurations of the multi-section arm, posed manually with all subsection markers visible and recorded in a
separate session; these configurations were not used to train $N_C$ or the multi-section curvature CNNs.
```

Response letter text (LaTeX) • snippets/resp_4.tex

```
We agree. For the multi-section arm, inter-section marker occlusion prevented tracking all backbone markers
during data collection, and the curvature labels were obtained from the motion-captured section-tip poses through
$N_C$. We therefore now describe the multi-section result as a model-constrained reconstruction and state its
assumptions explicitly in Section~2.2 and Section~4.2.2: planar bending within each section, no torsion, and a
bending direction that is constant along each section. The errors in Fig.~7b quantify the agreement with these
model-constrained labels. For Fig.~S2b, we now report the number of validation configurations (20, posed manually
with all subsection markers visible and recorded in a separate session), that the error bars are one standard
deviation across configurations, and that these configurations are independent of the data used to train $N_C$
(single-section dataset) and the three-section curvature estimator. We also report the error of $N_C$ on held-out
single-section test data ($N = 500$): 0.38--0.43~mm at $L/3$, 1.20--1.37~mm at $2L/3$, and 2.49--2.94~mm at $L$
for the three curvature models (SI Section~4.2).
```

Reviewer, Comment 8

[. . .] The authors should state: how samples were randomized; whether test data were separated by session or configuration neighborhood; whether controller targets and trajectories were excluded from training; and whether experiments span different days, camera settings, coverings, or robot builds.

Facts:

- The split is a random permutation of all pooled samples (seed 42), not blocked by session or neighborhood.
- The multi-section arm was logged at 1 Hz with each configuration held for ≈ 3 s, so 134 of the 500 test samples repeat a training configuration.
- The control targets were drawn from the sampled workspace.
- The acquisition days are in Table S12.

Step 63 Manuscript, Section 3.1.1: replace the split sentence.

Find: "The dataset was split into training (85%), validation (10%), and test (5%) sets."

LaTeX to paste • snippets/sec311_split.tex

```
The dataset was split into training (85\%), validation (10\%), and test (5\%) sets by a random permutation of all
samples with a fixed seed; the split was not blocked by acquisition session or by configuration neighborhood (see
Section~3.2 and SI Section~4.1).
```

Step 64 Manuscript, Section 3.2: insert after "... the final third utilizes full independent actuation."

LaTeX to paste • snippets/sec32_hold.tex

```
For the multi-section arm, each commanded configuration was held for about 3~s while samples were logged at 1~Hz,
so consecutive samples can share the same configuration.
```

Step 65 SI, end of the document: add Table S14 (effect of shared configurations).

LaTeX to paste • snippets/tab_S14_overlap.tex

```
\begin{table}[h]
\centering
\caption{Effect of configurations shared between training and test sets (multi-section arm). Because each
commanded configuration was held for about 3~s and logged at 1~Hz, 134 of the 500 test samples share their tendon
configuration with a training sample. Tip position errors [mm], mean (median).}
\label{tab:S14_overlap}
\begin{tabular}{lccc}
\toprule
Test samples & Tip I & Tip II & Tip III \\
\midrule
All (500) & 4.44 & 14.01 & 26.45 \\
Shared configuration (134) & 4.01 (3.54) & 12.07 (10.78) & 22.23 (18.36) \\
Other (366) & 4.60 (4.10) & 14.72 (12.79) & 27.99 (24.12) \\
\bottomrule
\end{tabular}
\end{table}
```

Preview (how it compiles)

Effect of configurations shared between training and test sets (multi-section arm). Because each commanded configuration was held for about 3 s and logged at 1 Hz, 134 of the 500 test samples share their tendon configuration with a training sample. Tip position errors [mm], mean (median).

Test samples	Tip I	Tip II	Tip III
All (500)	4.44	14.01	26.45
Shared configuration (134)	4.01 (3.54)	12.07 (10.78)	22.23 (18.36)
Other (366)	4.60 (4.10)	14.72 (12.79)	27.99 (24.12)

Response letter text (LaTeX) • `snippets/resp_8.tex`

```
We added this information to Section~3.1.1 and SI Section~4.1. All samples of each dataset were pooled and
randomly permuted with a fixed seed before the 85/10/5 division; the split was not blocked by session or by
configuration neighborhood. Because each commanded configuration of the multi-section arm was held for about 3~s
and recorded at 1~Hz, some test samples share their configuration with training samples. To quantify the effect,
we evaluated the test error with and without these samples: for the multi-section arm, the tip position errors
become 4.60, 14.72, and 27.99~mm (instead of 4.44, 14.01, and 26.45~mm) (Table~S14), and for the single-section
arm, samples close to a training configuration are not more accurate than the others (4.08 vs.\ 3.86~mm).
Session-blocked evaluation is listed as a limitation. The control targets were selected among the sampled
workspace configurations and were therefore not excluded from training; the closed-loop experiments use new
camera images, and the trajectories are not part of the datasets. The single-section data were collected in one
day and the multi-section data on two days, with the same camera settings and one build of each arm
\textbf{[CONFIRM]}; the two arms use different coverings (corrugated silicone and stretchable fabric,
Section~2.1).
```

Reviewer, SI comment 31

Tables S3 and S4 list dataset fields but not dataset composition. Add sample counts per acquisition mode and robot configuration, including how many samples were removed and why.

As agreed, Algorithm S1 stays as it is. The mode is drawn with equal probability for each configuration (`np.random.randint(0,3)`) and is not stored per sample, so the per-mode counts are given "by design".

Step 66 SI, Section 4.1: replace the section text (keep the heading).

Find: "The composition of the datasets for the single- and multi-segment manipulators is shown in Table S3 and Table S4, respectively. The sampled workspace of the multi-segment robot is shown in Fig. S1a, while the main reference systems of interest in Fig. S1b."

LaTeX to paste • `snippets/si41_composition.tex`

```
The composition of the datasets for the single- and multi-section manipulators is shown in Table~S3, Table~S4,
and Table~S12. The sampled workspace of the multi-section robot is shown in Fig.~S1a, and the main reference
systems in Fig.~S1b. The single-section dataset (10,000 samples) was recorded in one day by motor babbling with
tendon displacements sampled uniformly in $[-45, 10]$~mm. The multi-section dataset (10,000 samples) was recorded
on two days with the strategy of Algorithm~S1: for each commanded configuration, one of the three sampling modes
is selected with equal probability, so that each mode accounts for about one third of the configurations; the
mode was not stored for each sample. Each commanded configuration was held for about 3~s while samples were
logged at 1~Hz. The forward-kinematics dataset of $N_p^3$ (10,000 samples) was recorded in three sessions over
two further days. All samples of each dataset were pooled and randomly split (85/10/5\%, fixed seed).
\textbf{[FILL: Data were saved in batches that were inspected during acquisition; X batches (Y samples) were
discarded because ...]}

Because of this acquisition scheme, test samples can be close to training samples. For the single-section arm,
52.6\% of the test samples have a training sample within 1~mm in tendon space, but their error is not lower than
that of the others (4.08 vs.\ 3.86~mm). For the multi-section arm, 26.8\% of the test samples share their
configuration with a training sample; excluding them increases the tip position errors to 4.60, 14.72, and
27.99~mm, compared with 4.44, 14.01, and 26.45~mm on the full test set (Table~S14). The control targets were
selected among the sampled workspace points (Section~6.3) and were therefore not excluded from training; the
closed-loop experiments use new camera images, and the trajectories are not part of the datasets.
```

Step 67 SI, end of the document: add Table S12 (dataset composition).

LaTeX to paste • `snippets/tab_S12_composition.tex`

```
\begin{table}[h]
\centering
\caption{Dataset composition.}
\label{tab:S12_composition}
\begin{tabular}{>{\raggedright\arraybackslash}p{3.0cm} >{\raggedright\arraybackslash}p{3.6cm}
>{\raggedright\arraybackslash}p{3.6cm} >{\raggedright\arraybackslash}p{3.0cm}}
\toprule
 & Single-section & Multi-section & Forward kinematics ($N_p^3$) \\
```

```
\midrule
Samples & 10,000 & 10,000 & 10,000 \\
Acquisition days & 1 (2025-06-19) & 2 (2025-07-18, 2025-07-21) & 2 (2025-07-25/26), 3 sessions \\
Sampling & Motor babbling, $U(-45, 10)$~mm per tendon & Algorithm~S1, modes selected with equal probability;
independent mode: I $U(-25, 10)$, II $U(-35, 10)$, III $U(-40, 10)$~mm & \textbf{[FILL]} \\
Recorded tendon range [mm] & $-44.7$ to $12.7$ & $-60.6$ to $50.5$ & \textbf{[FILL]} \\
Logging & One sample per configuration & 1~Hz, configuration held $\approx 3$~s & \textbf{[FILL]} \\
Split (train/val/test) & 8,500 / 1,000 / 500 & 8,500 / 1,000 / 500 & 8,500 / 1,000 / 500 \\
Removed samples & \textbf{[FILL]} & \textbf{[FILL]} & \textbf{[FILL]} \\
\bottomrule
\end{tabular}
\end{table}
```

Preview (how it compiles)

Dataset composition.

	Single-section	Multi-section	Forward kinematics (N_p^3)
Samples	10,000	10,000	10,000
Acquisition days	1 (2025-06-19)	2 (2025-07-18, 2025-07-21)	2 (2025-07-25/26), 3 sessions
Sampling	Motor babbling, $U(-45, 10)$ mm per tendon	Algorithm S1, modes selected with equal probability; independent mode: I $U(-25, 10)$, II $U(-35, 10)$, III $U(-40, 10)$ mm	**[FILL]**
Recorded tendon range [mm]	-44.7 to 12.7	-60.6 to 50.5	**[FILL]**
Logging	One sample per configuration	1 Hz, configuration held ≈ 3 s	**[FILL]**
Split (train/val/test)	8,500 / 1,000 / 500	8,500 / 1,000 / 500	8,500 / 1,000 / 500
Removed samples	**[FILL]**	**[FILL]**	**[FILL]**

Response letter text (LaTeX) • `snippets/resp_31.tex`

```
We added the dataset composition (Table~S12 and SI Section~4.1): sample counts, acquisition days and sessions,
sampled and recorded tendon ranges, logging scheme, and split. For the multi-section arm, each commanded
configuration selected one of the three sampling modes of Algorithm~S1 with equal probability, so each mode
accounts for about one third of the configurations by design; the mode was not stored for each sample.
\textbf{[FILL: removed batches/samples and reason]}
```

3.6 Networks and training (Comments 9, 16, 17)

Reviewer, minor comments 16 and 17

Explain how angular sine/cosine outputs are converted back to angles and how losses for position, orientation, length, and curvature are balanced after normalization. Clarify the negative sign in Eq. S11 and clarify how the highlighted pixels are the regions used for prediction.

From the training notebooks:

- The networks output angles directly.
- Sine and cosine are applied to prediction and label inside the loss only.
- Positions, lengths and coefficients are standardized.
- All loss terms have equal weight.

Step 68 SI, Section 3.1: replace.

Find: "During training, outputs were normalized for stability, with all angular quantities reparameterized using sine and cosine."

LaTeX to paste • `snippets/si31_angles.tex`

```
During training, positions, lengths, and curvature coefficients were standardized with the mean and standard
deviation of the training set, and the networks regress the angles directly; in the loss, predicted and target
angles are compared through their sine and cosine to remove the $2\pi$ discontinuity, so no decoding step is
required at inference. All loss terms have equal weight.
```

Step 69 SI, Section 3.2: insert after Eq. S12 (before "Similar spatial attention results...").

LaTeX to paste • snippets/si32_gradcam.tex

```
The negative sign in Eq.~S11 makes $\alpha_k$ positive for channels whose activation reduces the loss, so that,
after the ReLU in Eq.~S12, the heatmap retains only the regions that contribute to lowering the prediction error.
Heatmaps are normalized to $[0, 1]$ per image and displayed without thresholding (Fig.~4a).
```

Response letter text (LaTeX) • snippets/resp_16.tex

```
The networks regress the angles directly; the sine and cosine of the predicted and target angles are only used
inside the loss to remove the $2\pi$ discontinuity, so no conversion back to angles is needed. Positions,
lengths, and curvature coefficients are standardized with the mean and standard deviation of the training set,
and all loss terms have equal weight. This is now stated in SI Section~3.1 and Table~S11.
```

Response letter text (LaTeX) • snippets/resp_17.tex

```
In Eq.~S11 the Grad-CAM score is the negative loss, so positive weights correspond to feature-map channels whose
activation reduces the prediction loss; after the ReLU of Eq.~S12, the heatmap retains only these regions. The
maps are normalized to $[0, 1]$ per image and shown without thresholding, so the color scale in Fig.~4a shows
this relative contribution. We clarified both points in SI Section~3.2 and in the caption of Fig.~4.
```

Reviewer, Comment 9

[. . .] camera model, lens and field of view, exposure and gain behavior, maximum epochs, random seeds, output normalization statistics, angular decoding, and the training details for N_C, N_p3, and N_l3. Additionally [. . .] tendon material and diameter, preload/zeroing criterion, current limit, sampled tendon ranges, and the number of acquisition sessions. The SI is a good place for a single reproducibility table containing these settings.

Step 70 Manuscript, Section 3.1.1: replace the optimizer sentence.

Find: "Training employed the Adam optimizer with $\eta = 10^{-3}$, weight decay, and early stopping."

LaTeX to paste • snippets/sec311_settings.tex

```
Training employed the Adam optimizer with $\eta = 10^{-3}$, weight decay, and early stopping; all training
settings are listed in Table~S11.
```

Step 71 SI, end of the document: add Table S11 (reproducibility) and fill the [FILL] cells.

LaTeX to paste • snippets/tab_S11_repro.tex

```
\begin{table*}[h]
\centering
\footnotesize
\caption{Reproducibility settings.}
\label{tab:S11_repro}
\begin{tabular}{>{\raggedright\arraybackslash}p{2.8cm} >{\raggedright\arraybackslash}p{6.2cm}
>{\raggedright\arraybackslash}p{6.2cm}}
\toprule
Item & Single-section & Multi-section \\
\midrule
Cameras & \textbf{[FILL: model, lens/field of view, size, mass]}; USB, OpenCV (V4L2), $640 \times 480$ grayscale
& Three cameras of the same model; one capture thread per camera, buffer of one frame, nominal 3~fps, latest
frame used (software synchronization) \\
Exposure and gain & \textbf{[FILL: automatic or fixed]} & \textbf{[FILL]} \\
Network input & $1 \times 480 \times 640$, 8-bit, scaled to $[0, 1]$ inside the network & $3 \times 480 \times
640$, one camera per channel \\
Pose CNN & MobileNetV2 (ImageNet weights), first convolution replaced by a 1-channel one; GAP (1280); linear head
$1280 \to 6$ & MobileNetV2 with 3-channel first convolution; GAP (1280); head $1280 \to 512$ (LeakyReLU) $\to 18$
\\
Curvature CNNs & Backbone initialized from $f_p^1$; linear heads $1280 \to 5 / 4 / 3$ & Backbone initialized from
$f_p^3$; heads $1280 \to 512 \to 15 / 12 / 9$ \\
Kinematic MLPs & $N_l^1$: $3 \to 128$ (linear, BatchNorm, LeakyReLU), three residual blocks of width 128, $128
\to 64 \to 32 \to 3$ & $N_p^3$: $9 \to 4 \times 128$ (ELU) $\to 6$; $N_l^3$: $6 \to 4 \times 128$ (ELU) $\to 9$,
trained through the frozen $N_p^3$ with $\zeta = 0.01$ on the standardized tendon vector \\
$N_C$ & $6 \to 128$, three residual blocks, $\to 5 / 4 / 3$; trained on single-section data & Applied to each
section, on the section-tip pose expressed in the frame of the preceding tip \\
Optimizer & Adam, learning rate $10^{-3}$, weight decay $5 \cdot 10^{-6}$ (CNNs) and $10^{-5}$ ($N_l^1$, $N_C$) &
Adam, learning rate $10^{-3}$, weight decay $5 \cdot 10^{-6}$ \\
Learning-rate schedule & Linear warm-up (10 iterations), then ReduceLROnPlateau (factor 0.5, patience 5) & Linear
warm-up (CNNs), then ReduceLROnPlateau (factor 0.6 for CNNs, 0.9 for MLPs; patience 5 and 15) \\
Epochs and early stopping & Maximum 500 epochs; patience 30 & Maximum 500 (CNNs), 800 ($N_p^3$), 2000 ($N_l^3$)
epochs; patience 50 \\
Batch size & 16 (images), 64 (MLPs) & 16 (images), 64 (MLPs); gradient clipping at 1.0 \\
Loss & \textbf{[FILL: Huber or MSE, see note]} & Huber, $\delta = 1$ \\
Angles & Regressed directly; sine and cosine inside the loss & Same \\
```

```
Output normalization & Mean and standard deviation of the training split, stored with the exported models & Same
\\
Augmentation & Gaussian blur (kernel 5, $\sigma \le 8$), Gaussian noise ($\sigma \le 30$), brightness and
contrast jitter (0.4), brightness shift (0.3), vignetting (0.6) & Same \\
Seed and split & 42 (Python, NumPy, PyTorch); 85/10/5\% random split of pooled samples & Same \\
Motors & Dynamixel XL330-M288-T, Protocol 2.0, 57,600 baud, extended position mode, pulley diameter 6~mm ($4096 /
(6\pi)$ ticks per mm), position tolerance 20 ticks & Same, nine motors; sections II and III actuated through
Bowden cables \\
Pretension and current limit & \textbf{[FILL: criterion]}; current limit \textbf{[FILL]}~mA; same procedure for
both arms & Same \\
Tendons & \textbf{[FILL: material, diameter]} & Same \\
Sampled tendon ranges & $U(-45, 10)$~mm per tendon & Independent mode: I $U(-25, 10)$, II $U(-35, 10)$, III
$U(-40, 10)$~mm; coupled modes through the Bending Solver \\
Acquisition sessions & One day (2025-06-19) & Two days (2025-07-18 and 2025-07-21); logging at 1~Hz,
configuration held for $\approx 3$~s; $N_p^3$ data: three sessions (2025-07-25/26) \\
Control parameters & $\lambda = 0.1$, $\varepsilon = 1$~mm, $\gamma = 3$~mm & $\lambda = 0.1$, $\varepsilon =
1$~mm, $\gamma = 10$ (SE(3) distance, Eq.~S16) \\
Control targets & 20, $k$-medoids on sampled positions with $z > 115$~mm & 20, same procedure with $z > 200$~mm
\\
\bottomrule
\end{tabular}
\end{table*}
```

Preview (how it compiles)

Reproducibility settings.

Item	Single-section	Multi-section
Cameras	**[FILL: model, lens/field of view, size, mass]**; USB, OpenCV (V4L2), 640×480 grayscale	Three cameras of the same model; one capture thread per camera, buffer of one frame, nominal 3 fps, latest frame used (software synchronization)
Exposure and gain	**[FILL: automatic or fixed]**	**[FILL]**
Network input	$1 \times 480 \times 640$, 8-bit, scaled to $[0, 1]$ inside the network	$3 \times 480 \times 640$, one camera per channel
Pose CNN	MobileNetV2 (ImageNet weights), first convolution replaced by a 1-channel one; GAP (1280); linear head $1280 \to 6$	MobileNetV2 with 3-channel first convolution; GAP (1280); head $1280 \to 512$ (LeakyReLU) $\to 18$
Curvature CNNs	Backbone initialized from f_p^1; linear heads $1280 \to 5/4/3$	Backbone initialized from f_p^3; heads $1280 \to 512 \to 15/12/9$
Kinematic MLPs	N_l^1: $3 \to 128$ (linear, BatchNorm, LeakyReLU), three residual blocks of width 128, $128 \to 64 \to 32 \to 3$	N_p^3: $9 \to 4 \times 128$ (ELU) $\to 6$; N_l^3: $6 \to 4 \times 128$ (ELU) $\to 9$, trained through the frozen N_p^3 with $\zeta = 0.01$ on the standardized tendon vector
N_C	$6 \to 128$, three residual blocks, $\to 5/4/3$; trained on single-section data	Applied to each section, on the section-tip pose expressed in the frame of the preceding tip
Optimizer	Adam, learning rate 10^{-3}, weight decay $5 \cdot 10^{-6}$ (CNNs) and 10^{-5} (N_l^1, N_C)	Adam, learning rate 10^{-3}, weight decay $5 \cdot 10^{-6}$
Learning-rate schedule	Linear warm-up (10 iterations), then ReduceLROnPlateau (factor 0.5, patience 5)	Linear warm-up (CNNs), then ReduceLROnPlateau (factor 0.6 for CNNs, 0.9 for MLPs; patience 5 and 15)
Epochs and early stopping	Maximum 500 epochs; patience 30	Maximum 500 (CNNs), 800 (N_p^3), 2000 (N_l^3) epochs; patience 50
Batch size	16 (images), 64 (MLPs)	16 (images), 64 (MLPs); gradient clipping at 1.0
Loss	**[FILL: Huber or MSE, see note]**	Huber, $\delta = 1$
Angles	Regressed directly; sine and cosine inside the loss	Same
Output normalization	Mean and standard deviation of the training split, stored with the exported models	Same
Augmentation	Gaussian blur (kernel 5, $\sigma \leq 8$), Gaussian noise ($\sigma \leq 30$), brightness and contrast jitter (0.4), brightness shift (0.3), vignetting (0.6)	Same
Seed and split	42 (Python, NumPy, PyTorch); 85/10/5% random split of pooled samples	Same
Motors	Dynamixel XL330-M288-T, Protocol 2.0, 57,600 baud, extended position mode, pulley diameter 6 mm ($4096/(6\pi)$ ticks per mm), position tolerance 20 ticks	Same, nine motors; sections II and III actuated through Bowden cables
Pretension and current limit	**[FILL: criterion]**; current limit **[FILL]** mA; same procedure for both arms	Same
Tendons	**[FILL: material, diameter]**	Same
Sampled tendon ranges	$U(-45, 10)$ mm per tendon	Independent mode: I $U(-25, 10)$, II $U(-35, 10)$, III $U(-40, 10)$ mm; coupled modes through the Bending Solver
Acquisition sessions	One day (2025-06-19)	Two days (2025-07-18 and 2025-07-21); logging at 1 Hz, configuration held for ≈ 3 s; N_p^3 data: three sessions (2025-07-25/26)
Control parameters	$\lambda = 0.1$, $\varepsilon = 1$ mm, $\gamma = 3$ mm	$\lambda = 0.1$, $\varepsilon = 1$ mm, $\gamma = 10$ (SE(3) distance, Eq. S16)
Control targets	20, k-medoids on sampled positions with $z > 115$ mm	20, same procedure with $z > 200$ mm

Step 72 Lighting claim: see Step 36.

Response letter text (LaTeX) • `snippets/resp_9.tex`

```
We added a reproducibility table to the SI (Table~S11) that collects the camera and acquisition settings, the network architectures, the training hyperparameters (optimizer, learning-rate schedule, maximum epochs, early stopping, batch size, random seed, normalization, angle handling, augmentation) for all networks including $N_C$, $N_p^3$, and $N_l^3$, the actuation hardware and pretensioning procedure, the tendon material and diameter, the sampled tendon ranges, and the acquisition sessions (Table~S12). We also limited the statements on lighting: image augmentations were used during training, but robustness to lighting or hardware variations was not tested, and this is now stated in Section~5.
```

3.7 Hardware and acquisition (Comments 13, 14, 15)

Reviewer, minor comments 13–15

The calibration paragraph says that 'all ten tendons' are tightened, but the multi-section arm has nine tendons and the single-section arm has three. Correct the number and state whether the same preload procedure is used for both systems. Identify the camera manufacturer and model, lens or field of view, nominal frame rate, and the physical size and mass of each camera and mount. The claim that images are acquired simultaneously should be supported by the synchronization method [. . .]

The paper actually says "all tendons"; the snippet states the numbers. In the code, the motors run in extended position mode (mode 4), and both acquisition scripts use the same pretension routine (`calibrate_motors`).

Step 73 Manuscript, Section 2.1, last paragraph: replace the calibration sentences.

Find: "Before each data acquisition session or experimental trial, the robots are calibrated by tightening all tendons through current control of the actuation motors. The motors then operate in position-control mode with a predefined current limit."

LaTeX to paste • `snippets/sec21_calibration.tex`

```
Before each data acquisition session or experimental trial, all tendons (three for the single-section arm and
nine for the multi-section arm) are pretensioned with the same procedure for both robots: \textbf{[FILL:
criterion, e.g., each motor is driven in current control until the tendon is taut at X~mA]}. The motors then
operate in extended position-control mode with a current limit of \textbf{[FILL]}~mA.
```

Step 74 Manuscript, Section 2.1, second paragraph: replace the first four sentences.

Find: "Internal cameras are added to each section for proprioception, and are rigidly mounted between sections and at the robot base. [. . .] at a resolution of 480×640 in grayscale."

LaTeX to paste • `snippets/sec21_cameras.tex`

```
Internal cameras (\textbf{[FILL: manufacturer and model]}; lens/field of view \textbf{[FILL]}; \textbf{[FILL:
size]}~mm, \textbf{[FILL: mass]}~g including the mount) are added to each section for proprioception, and are
rigidly mounted between sections and at the robot base. Consequently, each view is fixed relative to the base of
the subsequent section. The sensing setup has negligible impact on the compliance of the soft structure. Camera
images are transmitted to a laptop via three USB cables at a resolution of $480 \times 640$ in grayscale. Each
camera is read by its own software thread, which keeps only the most recent frame; at each control step, the
latest frames of the three cameras are used, so the images are acquired concurrently but are not
hardware-synchronized.
```

Response letter text (LaTeX) • `snippets/resp_13.tex`

```
We now state the number of tendons explicitly (three for the single-section arm and nine for the multi-section
arm) and that the same pretensioning procedure is used for both arms, including its criterion and the motor
current limit (Section~2.1 and Table~S11).
```

Response letter text (LaTeX) • `snippets/resp_14.tex`

```
The camera manufacturer and model, lens and field of view, nominal frame rate, and the size and mass of the
camera and mount are now given in Section~2.1 and in Table~S11.
```

Response letter text (LaTeX) • `snippets/resp_15.tex`

```
The acquisition is synchronized in software only: each camera is read by its own capture thread, and the
controller uses the latest frame of each camera. We removed the word ``simultaneously'' and describe this
mechanism in Section~2.1 and in the caption of Fig.~1.
```

3.8 Results and statistics (Comments 6, 7, 21, 24, 25; SI comment 29)

Reviewer, Comment 6

[. . .] H(l) penalizes displacement from the mean sampled tendon vector. It does not directly minimize tendon force, motor current, energy, or control effort. The objective should therefore might described as actuator-range centering unless effort is measured. For both controllers, an ablation with the secondary objective disabled using identical targets would be helpful. Results should report both primary-task error and a quantitative secondary-task metric.

There are two offline ablations, both labeled as simulations:

- Model-based: the existing `MULTI_nullspace_ablation`.

- Data-driven (new): N_l^3 retrained with $\zeta = 0.01$ and with $\zeta = 0$. The retrained $\zeta = 0.01$ network gives the same $H(\mathbf{l})$ on held-out poses as the deployed `IK_model.pt` (326 vs. 326), which validates the retraining.

Motor current is not logged, so no effort claim can be made.

Step 75 Manuscript, Section 3.3.2, sentence after Eq. 9: replace.

Find: "where $\bar{l}$ denotes the mean tendon actuation estimated from the dataset. Reducing the deviation from mid-range values keeps the manipulator away from its physical limits, preserving flexibility for future tasks, and promotes smooth, coordinated joint motions by distributing control effort evenly across the tendons."

LaTeX to paste • `snippets/sec332_centering.tex`

```
where $\bar{\mathbf{l}}$ denotes the mean tendon actuation estimated from the dataset. Reducing the deviation
from mid-range values keeps the tendon lengths away from their actuation limits (actuator-range centering) and
preserves flexibility for future tasks. Since motor currents were not recorded, we do not claim a reduction of
actuation effort.
```

Step 76 Manuscript, Section 3.3.2, after Eq. 11: replace.

Find: "The first term enforces task accuracy, while the second biases the solution toward $\bar{l}$. The secondary objective can be flexibly modified for alternative tasks such as obstacle avoidance or grasping."

LaTeX to paste • `snippets/sec332_eq11_after.tex`

```
The first term enforces task accuracy, while the second biases the solution toward $\bar{\mathbf{l}}$. Unlike the
null-space projection of Eq.~10, this penalty can trade some task accuracy for range centering (SI Section~5.3).
The secondary objective can be flexibly modified for alternative tasks such as obstacle avoidance or grasping.
```

Step 77 Manuscript, end of Section 4.2.4: insert.

LaTeX to paste • `snippets/sec425_ablation_pointer.tex`

```
An offline ablation of the secondary objectives of both controllers is reported in SI Section~5.3.
```

Step 78 SI, new Section 5.3 after Section 5.2: insert.

LaTeX to paste • `snippets/si53_ablation.tex`

```
\subsection{Secondary-Objective Ablation}
To isolate the effect of the secondary objectives, both controllers were evaluated offline, through their
kinematic models, on the 20 control targets of the multi-section arm (Fig.~S8, Table~S13). For the model-based
controller, the control law of Eq.~10 was replayed on the PCC model from the same initial state and target
sequence, with and without the null-space term. For the data-driven controller, $N_l^3$ was retrained with $\zeta
= 0.01$ (as in Section~3.3.2) and with $\zeta = 0$, with the same data, split, and seed, and evaluated through
the frozen $N_p^3$. In both cases the secondary objective reduces $H(\mathbf{l})$ and the deviation of the tendon
lengths from $\bar{\mathbf{l}}$. The null-space projection leaves the final position error essentially unchanged,
while the penalty in the training loss of $N_l^3$ slightly increases the position error predicted by $N_p^3$.
These results are simulated through the respective models and were not measured on the robot.
```

Step 79 SI, end of the document: add Table S13 and Fig. S8.

LaTeX to paste • `snippets/tab_S13_ablation.tex`

```
\begin{table}[h]
\centering
\caption{Offline ablation of the secondary objective on the 20 control targets of the multi-section arm (mean
$\pm$ standard deviation across targets). Position error: final error predicted by the model (PCC for the
model-based controller, $N_p^3$ for the data-driven one).}
\label{tab:S13_ablation}
\begin{tabular}{llccc}
\toprule
Controller & Secondary objective & $H(\mathbf{l})$ & RMS $|\mathbf{l} - \bar{\mathbf{l}}|$ [mm] & Position error
[mm] \\
\midrule
Model-based & with null-space term & $269 \pm 135$ & $7.48 \pm 1.96$ & $5.25 \pm 2.23$ \\
 & without & $693 \pm 396$ & $12.00 \pm 3.18$ & $5.84 \pm 1.85$ \\
\midrule
Data-driven & $\zeta = 0.01$ & $371 \pm 166$ & $8.84 \pm 2.07$ & $5.09 \pm 2.15$ \\
 & $\zeta = 0$ & $487 \pm 257$ & $10.07 \pm 2.61$ & $4.62 \pm 2.34$ \\
\bottomrule
\end{tabular}
\end{table}
```

Preview (how it compiles)

Offline ablation of the secondary objective on the 20 control targets of the multi-section arm (mean ± standard deviation across targets). Position error: final error predicted by the model (PCC for the model-based controller, N_p^3 for the data-driven one).

Controller	Secondary objective	$H(\mathbf{l})$	RMS $\lvert\mathbf{l}-\bar{\mathbf{l}}\rvert$ [mm]	Position error [mm]
Model-based	with null-space term	269 ± 135	7.48 ± 1.96	5.25 ± 2.23
	without	693 ± 396	12.00 ± 3.18	5.84 ± 1.85
Data-driven	$\zeta = 0.01$	371 ± 166	8.84 ± 2.07	5.09 ± 2.15
	$\zeta = 0$	487 ± 257	10.07 ± 2.61	4.62 ± 2.34

LaTeX to paste • snippets/figS8_ablation.tex

```
\begin{figure}[h]
\centering
\includegraphics[width=\linewidth]{nullspace_ablation.pdf}\\
\includegraphics[width=\linewidth]{ik_ablation.pdf}
\caption{Offline ablation of the secondary objective on the 20 control targets of the multi-section arm. (a) Model-based controller, Eq.~10 replayed on the PCC model with and without the null-space term. (b) Data-driven controller, $N_l^3$ trained with $\zeta = 0.01$ and $\zeta = 0$ and evaluated through the frozen $N_p^3$. Left to right: $H(\mathbf{l})$, RMS deviation of the tendon lengths from $\bar{\mathbf{l}}$, and final position error predicted by the model. Bars: mean; error bars: one standard deviation across targets.}
\label{fig:S8_ablation}
\end{figure}
```

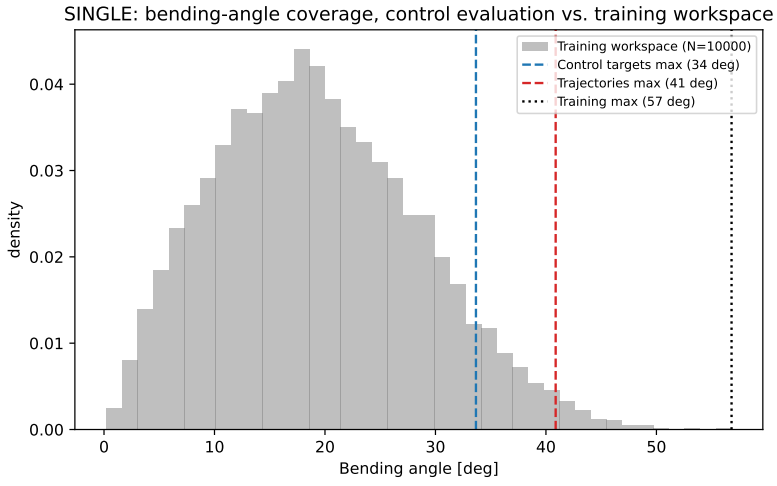
The effort wording of the Introduction and of Section 5 is handled in Step 23 and Step 38.

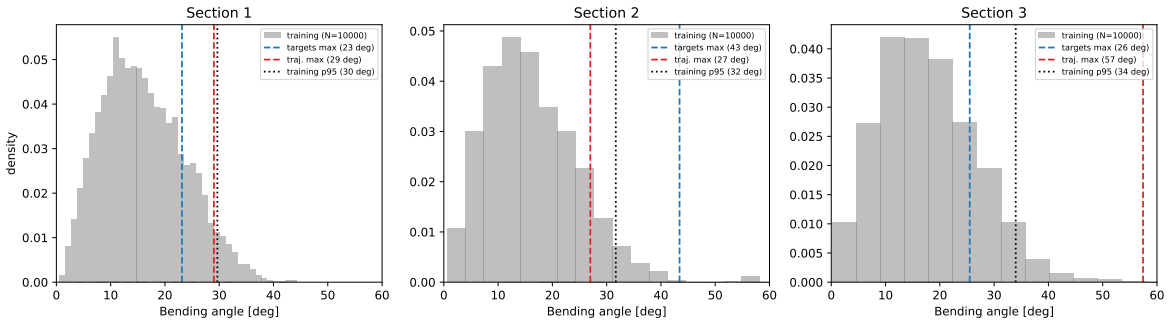
Response letter text (LaTeX) • snippets/resp_6.tex

```
We agree that $H(\mathbf{l})$ centers the tendon lengths in their actuation range and does not measure effort, which was not recorded. We now describe it as actuator-range centering and removed the statements about effort (Introduction, Section~3.3.2, Conclusion). To isolate its effect without new hardware experiments, we performed offline ablations on identical targets for both controllers (new SI Section~5.3, Fig.~S8, and Table~S13). For the model-based controller, replaying the control law on the PCC model for the 20 evaluation targets gives a final position error of $5.25 \pm 2.23$~mm with and $5.84 \pm 1.85$~mm without the null-space term, while the RMS deviation of the tendon lengths from $\bar{\mathbf{l}}$ decreases from $12.00 \pm 3.18$~mm to $7.48 \pm 1.96$~mm ($H(\mathbf{l})$: $693 \pm 396$ vs.\ $269 \pm 135$). For the data-driven controller, we retrained the inverse-kinematics network with and without the secondary term ($\zeta = 0.01$ and $\zeta = 0$, same data, split, and seed) and evaluated it through the frozen forward network on the same targets: the secondary term reduces $H(\mathbf{l})$ from $487 \pm 257$ to $371 \pm 166$ (RMS tendon deviation from 10.1 to 8.8~mm), at the cost of a small increase of the position error predicted by the forward network ($5.1 \pm 2.1$ vs.\ $4.6 \pm 2.3$~mm). Unlike the null-space projection, the penalty in the training loss trades some primary-task accuracy for range centering, which we now state in Section~3.3.2. These results are simulated through the respective models and are reported as such.
```

Reviewer, Comment 7

[...] The training datasets reportedly include bending angles up to 85° per section [...] The authors should report the target coordinates, associated bending and compression ranges, and the fraction of the sampled workspace represented by the evaluation set. Each of the 20 targets is visited only once. [...] Either additional load magnitudes and directions should be tested or the claims should be limited.





Step 80 Manuscript, Section 3.2: replace the bending range (measured values instead of 85°).

Find: "We collected 10,000 data points for both single- and multi- section configurations using six Optitrack Prime 13 cameras and Motive software, covering bending angles up to 85° per section."

LaTeX to paste • snippets/sec32_range.tex

```
We collected 10,000 data points for both single- and multi-section configurations using six Optitrack Prime 13
cameras and Motive software, covering bending angles up to $57^\circ$ for the single-section arm (95th percentile
$36^\circ$) and with 95th percentiles of $30^\circ$, $32^\circ$, and $34^\circ$ for sections I--III of the
multi-section arm.
```

Step 81 Manuscript, Section 4.1.4: replace the first sentences (this also adds the denominator).

Find: "Twenty workspace targets distributed via k-medoids clustering were reached sequentially by all controllers; each target was visited once, and errors are reported as the final position error at convergence. Results are shown in Fig. 5c."

LaTeX to paste • snippets/sec414_targets.tex

```
Twenty workspace targets distributed via $k$-medoids clustering were reached sequentially by all controllers;
each target was visited once, and errors are reported as the final position error at convergence, relative to the
150~mm section length for percentages. The standard deviations therefore describe the variation across targets,
not trial-to-trial repeatability. Results are shown in Fig.~5c.
```

Step 82 Manuscript, Section 4.2.6: replace the first sentence.

Find: "A 50 g payload (≈ 40% of effective robot mass, including TPU body, fabric, cameras, and PLA interfaces) was attached to the tip, emulating a sustained tip contact force and serving as a proxy for evaluating closed-loop robustness under persistent external interactions."

LaTeX to paste • snippets/sec426_payload.tex

```
A 50~g payload ($\approx 40\%$ of effective robot mass, including TPU body, fabric, cameras, and PLA interfaces)
was attached to the tip, emulating a sustained tip contact force. This single constant-load condition
demonstrates the compensation of this load by the closed-loop controller; it does not establish robustness to
general disturbances or contact.
```

Step 83 SI, Section 6.3: insert at the end of the section.

LaTeX to paste • snippets/si63_coverage.tex

```
Before clustering, the sampled workspace is restricted to points above $z = 115$~mm for the single-section arm
and $z = 200$~mm for the multi-section arm (89.9\% and 99.7\% of the samples), and each cluster center is
replaced by the nearest sampled point. The resulting targets are listed in Tables~S8 and S9. The convex hull of
the 20 targets covers 9.2\% (single section) and 16.4\% (multi-section) of the volume of the convex hull of the
sampled positions, and the median distance from a sampled position to the nearest target is 7.6~mm and 38.5~mm,
respectively. For the single-section arm, the targets reach a bending angle of $33.7^\circ$ and the trajectories
$40.9^\circ$, compared with a 95th percentile of $35.6^\circ$ in the training data (Fig.~S9).
```

Step 84 SI, end of the document: add Tables S8 and S9 and Fig. S9.

LaTeX to paste • snippets/tab_S8_targets_single.tex

```
\begin{table}[h]
\centering
\caption{Control targets of the single-section arm. Bending angle: PCC estimate from the target position.}
\label{tab:S8_targets_single}
\begin{tabular}{rrrrr}
\toprule
ID & $x$ [mm] & $y$ [mm] & $z$ [mm] & Bending [$^\circ$] \\
\midrule
```

```
1 & -19.8 & -25.4 & 122.6 & 29.3 \\
2 & -10.2 & -16.0 & 129.0 & 16.6 \\
3 & -7.3 & 32.8 & 121.8 & 30.7 \\
4 & -0.7 & -8.2 & 133.9 & 6.9 \\
5 & -1.1 & 18.3 & 128.2 & 16.1 \\
6 & 22.4 & 2.6 & 128.7 & 19.7 \\
7 & -24.8 & 5.9 & 123.2 & 23.2 \\
8 & 26.6 & -10.3 & 120.8 & 26.4 \\
9 & 12.8 & 14.2 & 125.8 & 17.1 \\
10 & -0.8 & -25.5 & 123.3 & 23.2 \\
11 & 27.7 & 18.1 & 120.8 & 30.5 \\
12 & 11.9 & -6.7 & 129.6 & 11.9 \\
13 & 13.6 & 28.8 & 122.8 & 29.0 \\
14 & 11.6 & -19.6 & 121.9 & 21.0 \\
15 & 5.6 & 6.5 & 135.8 & 7.2 \\
16 & 36.7 & 5.2 & 121.9 & 33.7 \\
17 & 2.4 & 2.0 & 122.8 & 2.8 \\
18 & -22.5 & -8.6 & 121.1 & 22.3 \\
19 & -19.6 & 19.5 & 122.6 & 25.2 \\
20 & -10.0 & 6.0 & 127.9 & 10.3 \\
\bottomrule
\end{tabular}
\end{table}
```

LaTeX to paste • snippets/tab_S9_targets_multi.tex

```
\begin{table*}[h]
\centering
\caption{Control targets of the multi-section arm (tip pose). Bending angle and length change of sections I / II / III: PCC estimate from the offline replay of the model-based controller.}
\label{tab:S9_targets_multi}
\begin{tabular}{rrrrrrrcc}
\toprule
ID & $x$ [mm] & $y$ [mm] & $z$ [mm] & Yaw [$^\circ$] & Pitch [$^\circ$] & Roll [$^\circ$] & Bending [$^\circ$] & $\Delta L$ [mm] \\
\midrule
1 & 82 & -167 & 308 & -15 & 21 & 62 & 13 / 31 / 26 & -17.1 / -15.4 / -13.7 \\
2 & -219 & 28 & 269 & 30 & -57 & 24 & 18 / 43 / 17 & -17.0 / -15.7 / -14.2 \\
3 & 129 & 103 & 332 & 13 & 25 & -37 & 14 / 25 / 16 & -17.6 / -15.8 / -14.3 \\
4 & 137 & 12 & 348 & -4 & 21 & 9 & 11 / 20 / 18 & -17.6 / -15.7 / -14.3 \\
5 & 4 & -123 & 353 & -0 & -2 & 39 & 6 / 21 / 21 & -17.5 / -15.6 / -14.2 \\
6 & 5 & -22 & 383 & 1 & 17 & 5 & 2 / 7 / 14 & -18.0 / -16.0 / -14.8 \\
7 & -72 & -72 & 366 & 6 & -34 & 26 & 6 / 17 / 15 & -17.2 / -15.3 / -14.0 \\
8 & -117 & -159 & 313 & 17 & -24 & 52 & 18 / 28 / 18 & -16.0 / -14.2 / -12.7 \\
9 & -68 & 141 & 344 & -6 & -21 & -47 & 15 / 23 / 1 & -17.0 / -15.2 / -14.0 \\
10 & 27 & 133 & 350 & 6 & 20 & -46 & 12 / 22 / 7 & -17.9 / -16.1 / -14.8 \\
11 & -140 & -17 & 352 & 5 & -38 & 11 & 11 / 22 / 11 & -16.8 / -14.9 / -13.7 \\
12 & -125 & 71 & 351 & 0 & -30 & -13 & 13 / 22 / 6 & -16.9 / -15.1 / -13.8 \\
13 & 81 & -56 & 359 & -10 & 37 & 31 & 4 / 19 / 22 & -18.5 / -16.6 / -15.2 \\
14 & -57 & 210 & 292 & -7 & -8 & -65 & 22 / 33 / 7 & -16.8 / -15.2 / -13.7 \\
15 & -179 & -66 & 308 & 9 & -56 & 19 & 15 / 34 / 18 & -17.0 / -15.4 / -13.9 \\
16 & 70 & 201 & 282 & 17 & 22 & -72 & 17 / 40 / 16 & -17.6 / -16.2 / -14.6 \\
17 & -68 & 7 & 379 & 4 & -11 & 9 & 6 / 12 / 9 & -17.6 / -15.6 / -14.4 \\
18 & -47 & 75 & 378 & 0 & -6 & -21 & 9 / 12 / 5 & -16.8 / -14.8 / -13.7 \\
19 & 52 & 48 & 376 & 3 & 2 & -8 & 7 / 9 / 11 & -18.2 / -16.2 / -15.0 \\
20 & -169 & 143 & 291 & -28 & -25 & -68 & 23 / 34 / 7 & -16.4 / -14.7 / -13.3 \\
\bottomrule
\end{tabular}
\end{table*}
```

LaTeX to paste • snippets/figS9.tex

```
\begin{figure}[h]
\centering
\includegraphics[width=0.7\linewidth]{single_workspace_coverage.pdf}\\
\includegraphics[width=\linewidth]{multi_workspace_coverage.pdf}
\caption{Bending-angle distribution of the training data compared with the control targets and trajectories. (a) Single-section arm. (b) Multi-section arm, sections I--III. Vertical lines mark the maximum bending of the targets, of the trajectories, and of the training data.}
\label{fig:S9_coverage}
\end{figure}
```

Preview (how it compiles)

Control targets of the single-section arm. Bending angle: PCC estimate from the target position.

ID	x [mm]	y [mm]	z [mm]	Bending [°]
1	-19.8	-25.4	122.6	29.3
2	-10.2	-16.0	129.0	16.6
3	-7.3	32.8	121.8	30.7
4	-0.7	-8.2	133.9	6.9
5	-1.1	18.3	128.2	16.1
6	22.4	2.6	128.7	19.7
7	-24.8	5.9	123.2	23.2
8	26.6	-10.3	120.8	26.4
9	12.8	14.2	125.8	17.1
10	-0.8	-25.5	123.3	23.2
11	27.7	18.1	120.8	30.5
12	11.9	-6.7	129.6	11.9
13	13.6	28.8	122.8	29.0
14	11.6	-19.6	121.9	21.0
15	5.6	6.5	135.8	7.2
16	36.7	5.2	121.9	33.7
17	2.4	2.0	122.8	2.8
18	-22.5	-8.6	121.1	22.3
19	-19.6	19.5	122.6	25.2
20	-10.0	6.0	127.9	10.3

Preview (how it compiles)

Control targets of the multi-section arm (tip pose). Bending angle and length change of sections I / II / III: PCC estimate from the offline replay of the model-based controller.

ID	x [mm]	y [mm]	z [mm]	Yaw [°]	Pitch [°]	Roll [°]	Bending [°]	ΔL [mm]
1	82	-167	308	-15	21	62	13 / 31 / 26	-17.1 / -15.4 / -13.7
2	-219	28	269	30	-57	24	18 / 43 / 17	-17.0 / -15.7 / -14.2
3	129	103	332	13	25	-37	14 / 25 / 16	-17.6 / -15.8 / -14.3
4	137	12	348	-4	21	9	11 / 20 / 18	-17.6 / -15.7 / -14.3
5	4	-123	353	-0	-2	39	6 / 21 / 21	-17.5 / -15.6 / -14.2
6	5	-22	383	1	17	5	2 / 7 / 14	-18.0 / -16.0 / -14.8
7	-72	-72	366	6	-34	26	6 / 17 / 15	-17.2 / -15.3 / -14.0
8	-117	-159	313	17	-24	52	18 / 28 / 18	-16.0 / -14.2 / -12.7
9	-68	141	344	-6	-21	-47	15 / 23 / 1	-17.0 / -15.2 / -14.0
10	27	133	350	6	20	-46	12 / 22 / 7	-17.9 / -16.1 / -14.8
11	-140	-17	352	5	-38	11	11 / 22 / 11	-16.8 / -14.9 / -13.7
12	-125	71	351	0	-30	-13	13 / 22 / 6	-16.9 / -15.1 / -13.8
13	81	-56	359	-10	37	31	4 / 19 / 22	-18.5 / -16.6 / -15.2
14	-57	210	292	-7	-8	-65	22 / 33 / 7	-16.8 / -15.2 / -13.7
15	-179	-66	308	9	-56	19	15 / 34 / 18	-17.0 / -15.4 / -13.9
16	70	201	282	17	22	-72	17 / 40 / 16	-17.6 / -16.2 / -14.6
17	-68	7	379	4	-11	9	6 / 12 / 9	-17.6 / -15.6 / -14.4
18	-47	75	378	0	-6	-21	9 / 12 / 5	-16.8 / -14.8 / -13.7
19	52	48	376	3	2	-8	7 / 9 / 11	-18.2 / -16.2 / -15.0
20	-169	143	291	-28	-25	-68	23 / 34 / 7	-16.4 / -14.7 / -13.3

The limitations in Section 5 are handled in Step 39.

Response letter text (LaTeX) • `snippets/resp_7.tex`

```
We thank the reviewer. We replaced the bending range in Section~3.2 with the measured one (single section:
maximum $57^\circ$, 95th percentile $36^\circ$; multi-section 95th percentiles of $30^\circ$, $32^\circ$, and
$34^\circ$), and we added the coordinates of all targets with their bending angle and length change (Tables~S8
and S9) and the coverage of the evaluation set (SI Section~6.3 and Fig.~S9). The single-section targets and
trajectories reach $33.7^\circ$ and $40.9^\circ$, close to the 95th percentile of the training data
($35.6^\circ$); the 20 targets span 9.2\% (single) and 16.4\% (multi) of the convex hull of the sampled
positions. We revised the text to state that the reported standard deviations describe the variation across
targets visited once, not repeatability, and that trajectory waypoints are serially correlated. The disturbance
experiment is now described as the compensation of a single constant 50~g off-center load rather than general
robustness. Repeated trials, non-planar and boundary trajectories, and further load conditions are listed as
limitations and future work in Section~5.
```

Reviewer, minor comments 21, 24, 25

State the denominator used for every percentage error. [. . .] In Table 1, the text states that orientation error remains below 9 degrees, but several standard deviations are larger than the corresponding means. Make clear that only the mean is below 9 degrees and report median, 95th percentile, and maximum error. The claim that bent configurations improve accuracy is plausible but currently qualitative. The authors should plot error against bending angle before drawing this conclusion.

Denominators, verified from the notebooks:

- Single section: 150 mm.
- Multi-section pose estimation: 150, 300 and 450 mm, from the base to the evaluated tip.
- Multi-section control: 450 mm.

Step 85 Manuscript, Section 4.1.1: replace the first result sentence.

Find: "The final position accuracy corresponds to $2.66 \pm 1.49\%$ of the robot length (a translation error of 3.99 ± 2.23 mm), and the overall rotation error is $3.11^\circ \pm 1.96^\circ$."

LaTeX to paste • snippets/sec411_pct.tex

```
The final position accuracy corresponds to $2.66 \pm 1.49\%$ of the robot length of 150~mm (a translation error of $3.99 \pm 2.23$~mm), and the overall rotation error is $3.11^\circ \pm 1.96^\circ$.
```

Step 86 Manuscript, Section 4.1.1: replace the bending sentence.

Find: "Overall, accuracy improves in bent configurations, where helicoidal deformations provide stronger visual cues for state estimation."

LaTeX to paste • snippets/sec411_bending.tex

```
Accuracy improves with bending (Spearman $\rho = -0.22$ between bending angle and position error, $p < 0.001$; SI Fig.~S7), consistent with helicoidal deformations providing stronger visual cues for state estimation.
```

Step 87 Manuscript, Section 4.2.1: replace the result sentences.

Find: "Position error for Module I ($2.96 \pm 1.79\%$ of section length) is consistent with the single-segment case. It accumulates along sections for Modules II and III, reaching $4.67 \pm 2.93\%$ and $5.88 \pm 3.66\%$ up to the respective tips, while orientation remains below 9°."

LaTeX to paste • snippets/sec421_pct.tex

```
Position error for Module I ($2.96 \pm 1.79\%$ of the 150~mm section length) is consistent with the single-section case. It accumulates along sections for Modules II and III, reaching $4.67 \pm 2.93\%$ and $5.88 \pm 3.66\%$ of the length from the base to the respective tips (300 and 450~mm), while the mean orientation error remains below $9^\circ$ (median below $7^\circ$; Table~1). The large standard deviations of the orientation error are caused by 10 of the 500 test samples (2\%), most of which have motion-capture orientation labels inconsistent with the sampled range (SI Section~6.1).
```

Step 88 Manuscript, Table 1: replace (means as printed in the paper, plus median, p95 and max).

LaTeX to paste • snippets/tab1.tex

```
\begin{table}[h]
\centering
\caption{Pose errors for the multi-section arm on the test set ($N = 500$).}
\label{tab:1}
\begin{tabular}{lcccc|cccc}
\toprule
 & \multicolumn{4}{c|}{Position [mm]} & \multicolumn{4}{c}{Orientation [$^\circ$]} \\
Section & mean $\pm$ std & median & p95 & max & mean $\pm$ std & median & p95 & max \\
\midrule
I & $4.44 \pm 2.68$ & 3.92 & 8.98 & 18.98 & $6.78 \pm 15.94$ & 4.77 & 10.18 & 176.71 \\
II & $14.01 \pm 8.79$ & 11.79 & 29.83 & 66.00 & $6.45 \pm 3.74$ & 5.69 & 13.31 & 28.21 \\
III & $26.45 \pm 16.49$ & 22.60 & 56.55 & 131.26 & $8.74 \pm 13.05$ & 6.94 & 16.81 & 176.78 \\
\bottomrule
\end{tabular}
\end{table}
```

Preview (how it compiles)

Pose errors for the multi-section arm on the test set ($N = 500$).

	Position [mm]				Orientation [°]			
Section	mean ± std	median	p95	max	mean ± std	median	p95	max
I	4.44 ± 2.68	3.92	8.98	18.98	6.78 ± 15.94	4.77	10.18	176.71
II	14.01 ± 8.79	11.79	29.83	66.00	6.45 ± 3.74	5.69	13.31	28.21
III	26.45 ± 16.49	22.60	56.55	131.26	8.74 ± 13.05	6.94	16.81	176.78

Step 89 Manuscript, Section 4.2.1, last sentence: replace (the data do not support it for the multi-section arm).

Find: "Bent configurations, however, partially offset accumulation by providing more distinctive image features."

LaTeX to paste • snippets/sec421_bending.tex

```
For the multi-section arm, the tip errors show no clear dependence on the bending angle of the sections (SI
Fig.~S7).
```

Step 90 Manuscript, Section 4.2.4: replace (adds the denominator).

Find: "Closed-loop controllers achieve near-sensing accuracy; the model-based controller reaches $5.6 \pm 2.5\%$ position error and the data-driven one $6.1 \pm 2.6\%$, with differences linked to whether the Jacobian uses encoder or camera feedback."

LaTeX to paste • snippets/sec424_pct.tex

```
Closed-loop controllers achieve near-sensing accuracy; the model-based controller reaches $5.6 \pm 2.5\%$
position error (relative to the 450~mm arm length) and the data-driven one $6.1 \pm 2.6\%$, with differences
linked to whether the Jacobian uses encoder or camera feedback.
```

Step 91 Manuscript, Section 4.2.3: add the denominator of "$5.1 \pm 3.3\%$ position" and "$2.1 \pm 1.2\%$ translation".

[FILL: e.g. "of the 450 mm arm length"; check the forward-kinematics notebook]

Step 92 SI, new Section 6.6 after Section 6.5: insert. Fig. S7 goes at the end of the SI.

LaTeX to paste • snippets/si_bending.tex

```
\subsection{Error as a Function of the Bending Angle}
Fig.~S7 relates the tip position error on the test sets to the bending angle. For the single-section arm, the
error decreases with bending (Spearman $\rho = -0.22$, $p < 0.001$ for position; $\rho = -0.14$, $p = 0.001$ for
orientation). For the multi-section arm, the tip errors show no clear dependence on the bending angle of the
corresponding section ($|\rho| \le 0.06$).
```

LaTeX to paste • snippets/figS7_bending.tex

```
\begin{figure}[h]
\centering
\includegraphics[width=\linewidth]{bending_error.pdf}
\caption{Tip position error as a function of the bending angle on the test sets ($N = 500$). (a) Single-section
arm; bending angle from the tip orientation label. (b--d) Multi-section arm, tips I--III, against the bending
angle of the corresponding section, computed from the relative orientation of consecutive tips. Dots: test
samples; black line: median in $5^\circ$ bins; shaded band: interquartile range; $\rho$: Spearman rank
correlation. Samples with a labeled section bending above $60^\circ$ (Section~6.1) lie outside the plotted
range.}
\label{fig:S7_bending}
\end{figure}
```

Response letter text (LaTeX) • snippets/resp_21.tex

```
We now state the normalization of every percentage at its first use: the section length (150~mm) for the
single-section arm; for the multi-section arm, the length from the base to the evaluated tip (150, 300, and
450~mm) for the pose estimation, and the 450~mm arm length for the control experiments.
```

Response letter text (LaTeX) • snippets/resp_24.tex

```
Table~1 now also reports the median, 95th percentile, and maximum error, and the text states that the mean
orientation error is below $9^\circ$ (the median is below $7^\circ$). The large standard deviations and maxima
are caused by 10 of the 500 test samples (2\%), most of which have motion-capture orientation labels inconsistent
with the sampled range; we discuss them in SI Section~6.1 (see also our response to the comment on Table~S6).
```

Response letter text (LaTeX) • snippets/resp_25.tex

```
We plotted the error against the bending angle (new Fig.~S7). For the single-section arm, the error decreases
with bending (Spearman $\rho = -0.22$, $p < 0.001$), which supports the statement in Section~4.1.1, now given
with this value. For the multi-section arm, no clear dependence was found, and we revised the corresponding
sentence in Section~4.2.1.
```

Reviewer, SI comment 29

Table S6 contains highly skewed errors, particularly 11.92 +/- 25.10 mm for Section II and orientation standard deviations near 16 degrees. Report robust distribution summaries and discuss the outliers rather than relying only on mean and standard deviation.

TABLE S6 | Relative pose errors for the multi-section arm.

Section	Position [mm]	Orientation [°]
I	4.44 ± 2.68	6.78 ± 15.94
II	11.92 ± 25.10	7.16 ± 16.02
III	8.86 ± 5.50	7.00 ± 13.17

Step 93 SI, Table S6: replace.

Note: the Section II orientation is 7.66 ± 16.02. The paper prints 7.16, which is a typo; the snippet uses 7.66.

LaTeX to paste • `snippets/tabS6.tex`

```
\begin{table*}[h]
\centering
\caption{Relative pose errors for the multi-section arm on the test set ($N = 500$), with robust summaries
(median [interquartile range], 95th percentile, maximum).}
\label{tab:S6}
\begin{tabular}{lcccc|cccc}
\toprule
 & \multicolumn{4}{c|}{Position [mm]} & \multicolumn{4}{c}{Orientation [$^\circ$]} \\
Section & mean $\pm$ std & median [IQR] & p95 & max & mean $\pm$ std & median [IQR] & p95 & max \\
\midrule
I & $4.44 \pm 2.68$ & 3.92 [2.60, 5.83] & 8.98 & 18.98 & $6.78 \pm 15.94$ & 4.77 [3.26, 6.64] & 10.18 & 176.71 \\
II & $11.92 \pm 25.10$ & 8.57 [5.79, 12.46] & 18.53 & 276.46 & $7.66 \pm 16.02$ & 5.59 [3.85, 7.76] & 11.78 &
177.07 \\
III & $8.86 \pm 5.50$ & 7.79 [5.34, 11.45] & 17.46 & 67.30 & $7.00 \pm 13.17$ & 5.50 [3.34, 7.70] & 12.68 &
174.14 \\
\bottomrule
\end{tabular}
\end{table*}
```

Preview (how it compiles)

Relative pose errors for the multi-section arm on the test set ($N = 500$), with robust summaries (median [interquartile range], 95th percentile, maximum).

	Position [mm]				Orientation [°]			
Section	mean ± std	median [IQR]	p95	max	mean ± std	median [IQR]	p95	max
I	4.44 ± 2.68	3.92 [2.60, 5.83]	8.98	18.98	6.78 ± 15.94	4.77 [3.26, 6.64]	10.18	176.71
II	11.92 ± 25.10	8.57 [5.79, 12.46]	18.53	276.46	7.66 ± 16.02	5.59 [3.85, 7.76]	11.78	177.07
III	8.86 ± 5.50	7.79 [5.34, 11.45]	17.46	67.30	7.00 ± 13.17	5.50 [3.34, 7.70]	12.68	174.14

Step 94 SI, Section 6.1: insert at the end of the section.

LaTeX to paste • `snippets/si61_outliers.tex`

```
The distributions in Table~1 and Table~S6 are skewed by a small number of samples. Ten of the 500 test samples
(2\%) have an orientation error above $30^\circ$ at one or more tips. Nine of them form three groups of three
consecutive samples, i.e., three commanded configurations each recorded three times, and their motion-capture
orientation labels imply section bending angles of $87^\circ$--$174^\circ$, far beyond the sampled range (95th
percentile $\approx 30^\circ$--$34^\circ$); this is consistent with orientation flips of the tracked rigid bodies
rather than with errors of the network. Excluding these 10 samples, the relative position error of Section~II is
$9.22 \pm 4.68$~mm (instead of $11.92 \pm 25.10$~mm) and its orientation error $5.97^\circ \pm 3.03^\circ$, and
the orientation error of Section~I is $5.10^\circ \pm 2.70^\circ$ (instead of $6.78^\circ \pm 15.94^\circ$).
```

Response letter text (LaTeX) • `snippets/resp_29.tex`

```
We added the median, interquartile range, 95th percentile, and maximum to Table~S6 (and Table~1). The large
standard deviations are caused by 10 of the 500 test samples (2\%): nine of them are three configurations
recorded three times each, whose motion-capture orientation labels imply section bending angles of
$87^\circ$--$174^\circ$, far outside the sampled range, which is consistent with orientation flips of the tracked
rigid bodies. Excluding these samples, the Section~II relative position error is $9.22 \pm 4.68$~mm instead of
$11.92 \pm 25.10$~mm. We discuss these outliers in SI Section~6.1 and report both summaries.
```

3.9 Remaining responses

The opening of the letter (overall assessment) and the responses whose changes are all in Part 1.

Response letter text (LaTeX) • `snippets/resp_overall.tex`

```
We thank the reviewer for the detailed and constructive assessment. We agree that the contribution lies in the integration of distributed internal cameras, markerless at inference, with model-based and learned redundancy-aware control on a multi-section arm, and we have revised the manuscript to state this precisely and to align the claims with the evidence. In summary: (i) we added the suggested references, a comparison paragraph in the Introduction, and a comparison table in the SI; (ii) we state explicitly that the curvature reconstruction does not enter the control laws and explain the encoder offset in Algorithm~1; (iii) we narrowed the scalability claim to networks retrained with a transferred backbone, validated for one and three sections; (iv) we describe the multi-section curvature result as a model-constrained reconstruction and state its assumptions; (v) we added an offline ablation of the secondary objectives of both controllers; (vi) we quantified the coverage of the control evaluation and limited the claims on repeatability and disturbances; (vii) we documented the data split and its effect, and added a reproducibility table; and (viii) we revised the figures and captions as requested. Our responses to each comment follow; changes in the manuscript are highlighted.
```

Response letter text (LaTeX) • `snippets/resp_10.tex`

```
We revised the figures and captions. Figure~3 has been redrawn with a single left-to-right flow, the name of each network on its path, the inputs and outputs of every network with tensor dimensions, and colors that separate pose estimation, curvature reconstruction, forward kinematics, and inverse kinematics; the caption now lists the inputs and outputs of all networks and states which architecture is used for the single- and multi-section kinematic networks. The captions of Figs.~1 and 2 now describe every labeled element (including the boundary arcs $L_1$--$L_4$, which are geometric constructions and not tendons), and Fig.~5b now uses equal axis scaling.
```

Response letter text (LaTeX) • `snippets/resp_22.tex`

```
We agree that the non-uniform scaling distorted the shapes. The shape panels of Fig.~5b now use equal axis scaling, and the caption no longer refers to non-uniform scaling.
```

Response letter text (LaTeX) • `snippets/resp_23.tex`

```
We agree that the summed statistic was ambiguous and not comparable between the two arms. Figure~7 now shows the 3D error at each reference point against the normalized arc length, for both arms on the same horizontal axis (median and interquartile range over the 500 test samples), and the caption and Sections~4.1.2 and 4.2.2 define the statistic.
```

Response letter text (LaTeX) • `snippets/resp_26.tex`

```
The captions of Figs.~6, 9, and 10 now state that each path is a single run of one counter-clockwise cycle starting at $(r, 0)$, and a new table (Table~S7) reports the radial and height RMS deviation from the reference circle for every controller, arm, and load condition. For example, under the 50~g payload the radial RMS error on the 120~mm circle is 11.46~mm in closed loop and 27.10~mm in open loop.
```

Response letter text (LaTeX) • `snippets/resp_28.tex`

```
The caption of Fig.~S2 and SI Section~4.2 now report the number of validation configurations (20), how they were obtained (posed manually with all markers visible, in a separate session not used for training), and that the error bars are one standard deviation across configurations. We also report the error of $N_C$ for each curvature model on held-out single-section test data (please see our response to Comment~4).
```

Response letter text (LaTeX) • `snippets/resp_30.tex`

```
Figures~S4--S6 now label each of the 20 targets with the same identifier in every panel, and use a single color scale per quantity within each figure. The target coordinates are listed with the same identifiers in Tables~S8 and S9.
```

3.10 Information only you can provide

- Camera: manufacturer, model, lens/field of view, size, mass, mount; exposure and gain (automatic or fixed). Snippets: `sec21_cameras`, `tab_S11_repro`.
- Pretension criterion and motor current limit. Snippets: `sec21_calibration`, `tab_S11_repro`.
- Tendon material and diameter. Snippet: `tab_S11_repro`.
- Single-section training loss (Huber or MSE, note 4). Snippet: `tab_S11_repro`.
- Removed batches/samples and why. Snippets: `si41_composition`, `tab_S12_composition`, `resp_31`.
- Forward-kinematics dataset: sampling, recorded range and logging (`tab_S12_composition`); the denominator in Section 4.2.3.
- Confirm one build per arm (`resp_8`) and S_{I}–S_{III} in Fig. 1b; check the cells listed in Section 2; give your BibTeX key for Ref. [15].

3.11 Notes for you only

For you only (not for the reviewer)

1. **Fig. S2b is computed with a parsing bug.**
   - *Cause.* `MULTI_helyx_testing.ipynb`, code cell 26, reads `CurvatureDataCheck.npz` with `np.hsplit(data, 5)`. That assumes the columns are grouped by parameter, but `MULTI_curvature_checks.py` (l. 594) writes them grouped by section (`[S1, c01, c11, c21, phi1, S2, ...]`). Every parameter lands in the wrong slot, and the backbones built by that cell are 1.7–3.8 mm long instead of ≈ 376 mm.
   - *Evidence.* This parsing reproduces exactly the published bars (6.4, 6.7, 46.5±21.5 mm). Read correctly, the N_C error on the 20 configurations is 214±198, 240±205 and 205±214 mm (sum over 9 points), the same order as Fig. S2a.
   - *Consequence.* The SI sentence "the error is much smaller than the one in Fig. S2a" is not supported.
   - *Handling, as you decided.* The paper is not changed on this point, and no snippet quotes values of S2b. The S2b caption snippet only describes how the figure is computed, which is true either way. Note that the notebook is in the public repository.
2. Direct comparison with markers, for your information:
   - Single section: the CNN backbone is 1.6–1.8 / 7.9–8.2 / 5.7–5.8 mm from the subsection markers at $L/3$, $2L/3$, L; the fitted labels themselves are 1.7 / 8.1 / 3.5 mm from the markers.
   - Multi-section: the CNN section ends are 9 / 32 / 65 mm from the motion-capture tips (labels: 5.5 / 23 / 50 mm), against 4.4 / 14.0 / 26.4 mm for the pose network.
3. Table S6, Section II orientation: 7.66 ± 16.02 (notebook output and recomputation); the paper prints 7.16. The new Table S6 snippet uses 7.66.
4. Loss: the multi-section notebooks use Huber ($\delta = 1$), but the single-section training notebook uses `nn.MSELoss()` and the Grad-CAM target in the code is $-$MSE. The paper says Huber for all regressors and in Eq. S11. Decide what to write in Table S11.
5. Fig. 3b and Section 3.1: the residual MLP is the architecture of N_l^1 and N_C; N_p^3 and N_l^3 are plain ELU MLPs. The snippets of Fig. 3 and Section 3.1 already say this.
6. Single-section circles: the logged targets have radii 31.4 and 22.4 mm (model-based runs) and 34.6 and 24.9 mm (data-driven runs), while Section 4.1.5 says 35 and 25 mm for all. Table S7 does not list radii, so it does not contradict the text.
7. Target selection: SI 6.3 says "iterative k-medoids", but the code runs k-means after the z filter and snaps each center to the nearest sampled point. The coverage snippet describes the snapping and the filter without naming the algorithm.
8. Section 4.2.6: $64.56/450 = 14.3\%$, but the paper prints $14.5 \pm 3.5\%$ for the open-loop translation error under load. Check.
9. The repository notebooks `SINGLE_test_reconstruction.ipynb` and `MULTI_test_reconstruction.ipynb` split with `random_state=42` per call. That gives the same training set but a different validation/test partition, so their numbers differ slightly from the paper. Their markdown says the original split is not reproducible; it is (global seed).
10. `REVIEW_TO_DELETE/` and `.claude/` are removed from the working tree (staged, not committed), but they remain in the public Git history.

3.12 Final steps

Step 95 SI: add a section "Additional Tables and Figures" at the end, and paste these snippets in this order (all shown above).

1. Tables: `tab_S7_paths`, `tab_S8_targets_single`, `tab_S9_targets_multi`, `tab_S10_comparison`, `tab_S11_repro`, `tab_S12_composition`, `tab_S13_ablation`, `tab_S14_overlap`.
2. Figures: `figS7_bending`, `figS8_ablation`, `figS9`.
3. Copy `figures_for_SI/*.pdf` next to the SI source.
4. The SI needs `booktabs` and `array` (or replace the rules with `\hline`) and the four new bibliography entries.

Step 96 Response letter: compile `snippets/response_letter.tex`, fill the `[FILL]` and `[CONFIRM]` items, and adjust the tone.